

Image Segmentation - 1

Use of segmentation

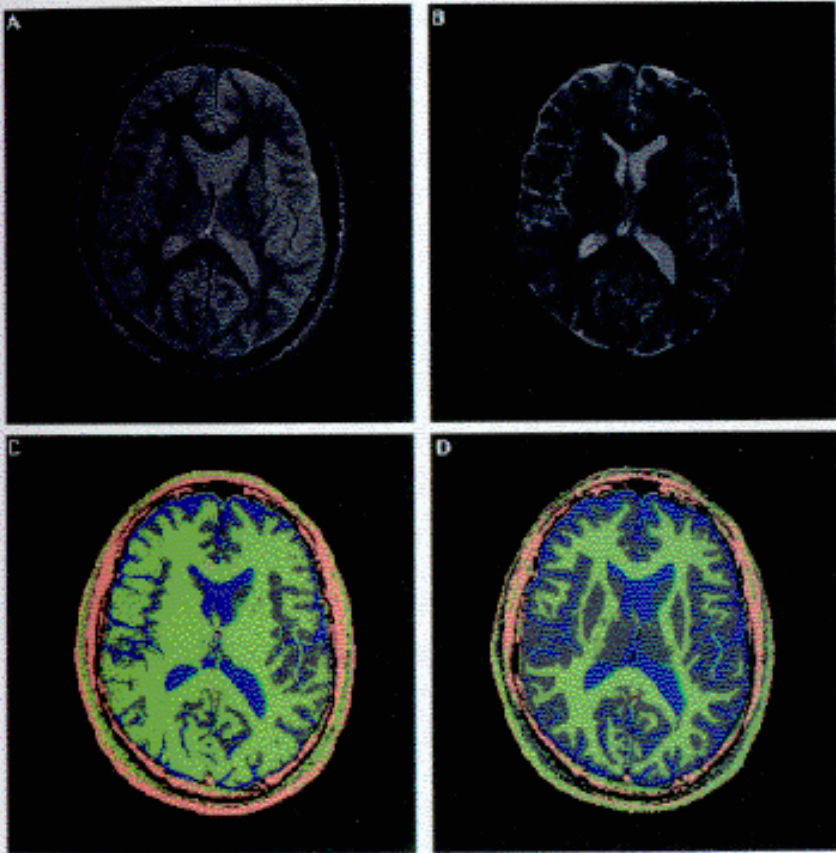


PLATE 4 The results of adaptive segmentation applied to dual-echo images of the brain. (A) Original T2-weighted image, (B) original proton-density weighted image, (C) result of conventional statistical classification, (D) result of EM segmentation. The tissue classes are represented by colors: blue, CSF; green, white matter; gray, gray matter; pink, fat; black, background. (Courtesy of Dr. W. M. Wells III, Surgical Planning Lab, Department of Radiology, Brigham and Women's Hospital, Boston.)

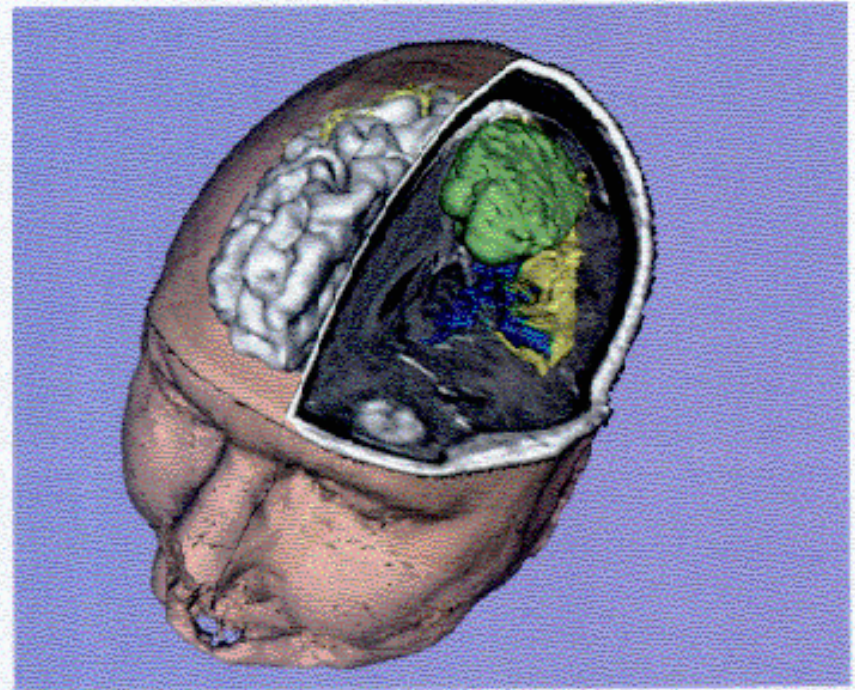


PLATE 6 Rendering of 3D anatomical models and 2D MRI cross-sections of a patient with a meningioma. The models of the skin surface, the brain, and the tumor (green) are based on automatically segmented 3D MRI data. The precentral gyrus (yellow) and the corticospinal tract (blue) are based on a previously aligned digital brain atlas [61]. (Courtesy of Drs. Ron Kikinis, Michael Kaus, and Simon Warfield, Surgical Planning Lab, Department of Radiology, Brigham and Women's Hospital, Boston.)

Image Segmentation

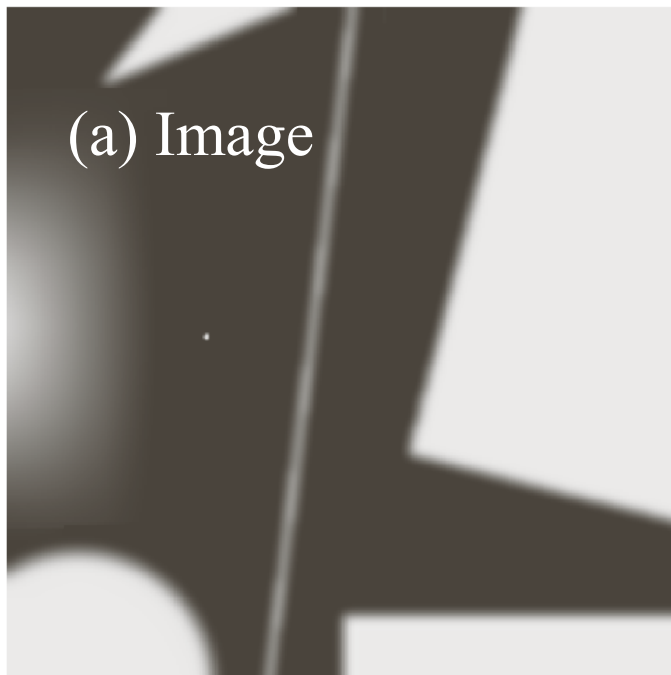
<http://en.wikipedia.org/wiki/Segmentation> *(image processing)*

1. Segmentation subdivides an image into its constituent regions (each region can represent an object).
2. Segmentation accuracy determines the eventual success or failure of computerized analysis procedures.
3. In general, image segmentation algorithms are based on intensity (a) discontinuity and (b) similarity.
4. Discontinuity in intensity value (sudden/abrupt change of intensity values), e.g., edges.
5. Similarity in intensity value, e.g., regions.

Gradient based segmentation

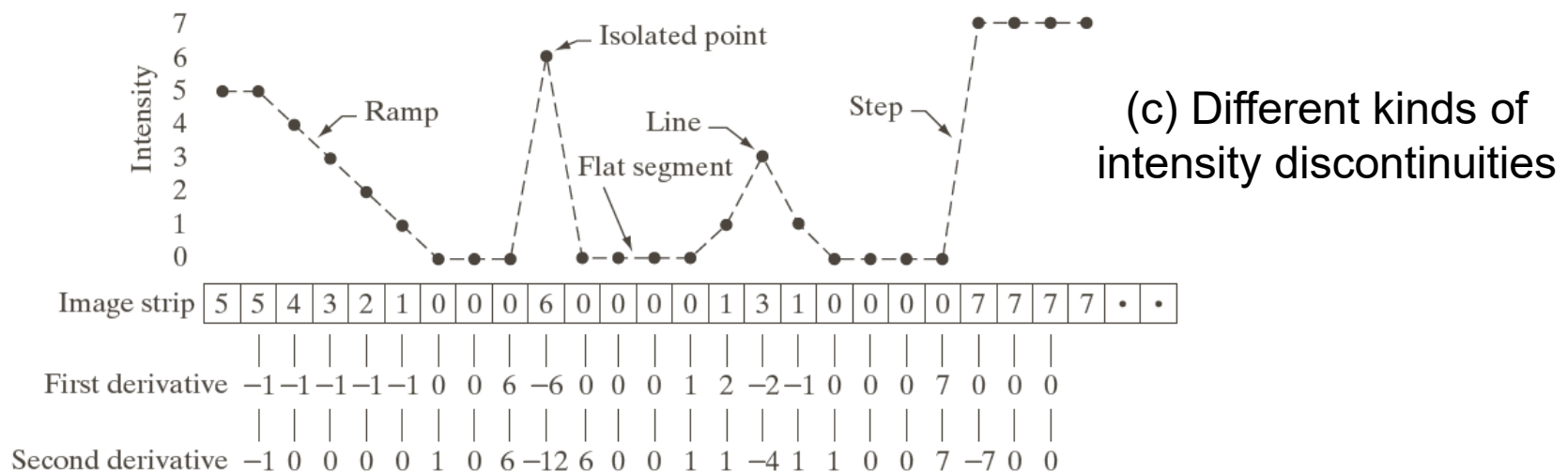
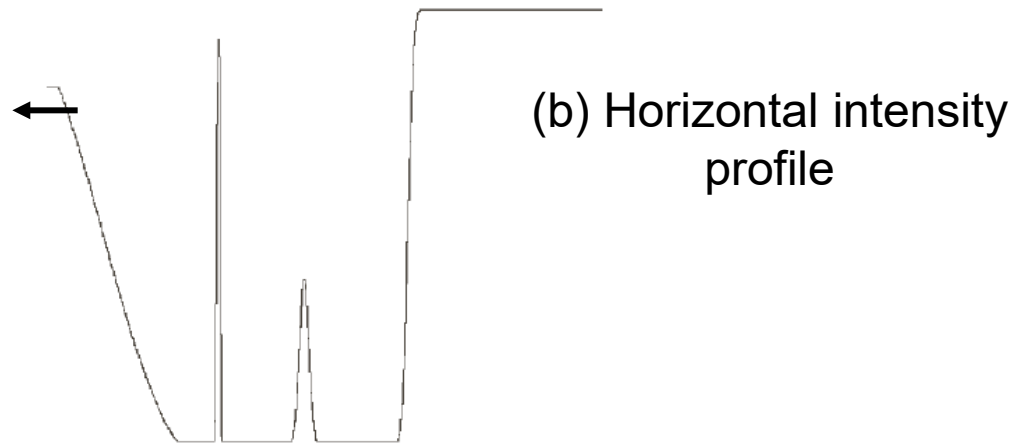
1. Discontinuity = large gradient magnitude = edge of the boundary
2. Boundary detection for segmentation based on detected edge elements
 - a. (Local) Edge linking based on edge strength, direction, and distance
 - b. Hough transforms

Where are intensity discontinuities?



a b
c

FIGURE 10.2 (a) Image. (b) Horizontal intensity profile through the center of the image, including the isolated noise point. (c) Simplified profile (the points are joined by dashes for clarity). The image strip corresponds to the intensity profile, and the numbers in the boxes are the intensity values of the dots shown in the profile. The derivatives were obtained using Eqs. (10.2-1) and (10.2-2).



Detection of Intensity Discontinuities

1. There are three different types of intensity discontinuities in a digital image:
 - a. Point (Isolated Point)
 - b. Line
 - c. Edge (Ideal, Ramp and Roof)
2. Intensity discontinuity is detected based on the mask response R within a pre-defined window, e.g., 3x3 window.

w_1	w_2	w_3
w_4	w_5	w_6
w_7	w_8	w_9

$$\text{Mask Response } R = \sum_{i=1}^9 w_i z_i$$

where w_i represent weights within a pre-defined window; z_i represent intensity values.

FIGURE 10.3
A general 3×3
spatial filter mask.

An image is represented
by intensity values z_i



w_i represent weights within
a 3x3 pre-defined window

w_1	w_2	w_3
w_4	w_5	w_6
w_7	w_8	w_9

Mask response at a pixel, $R = \sum_{i=1}^9 w_i z_i$

Working principle: For each pixel, x , in an image, if the absolute value of the mask response, $|R|$, is larger than or equal to some threshold T , $|R| \geq T$, then an intensity discontinuity is detected and located at the pixel.

Point Detection

1. If $|R| \geq T$, then a point has been detected. This point is the location on which the mask is centred, where T is a non-negative threshold.
2. For example, an isolated point is detected if it is different from its surroundings.

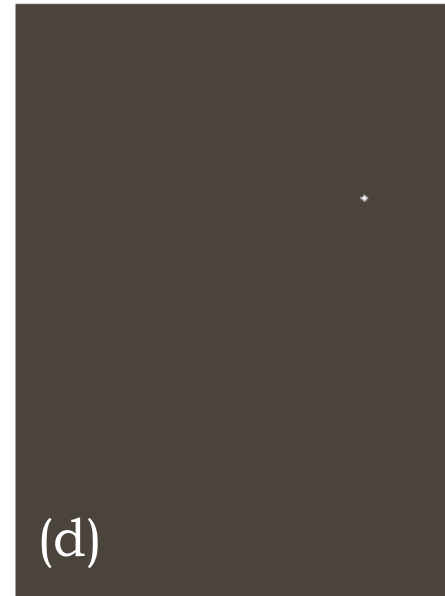
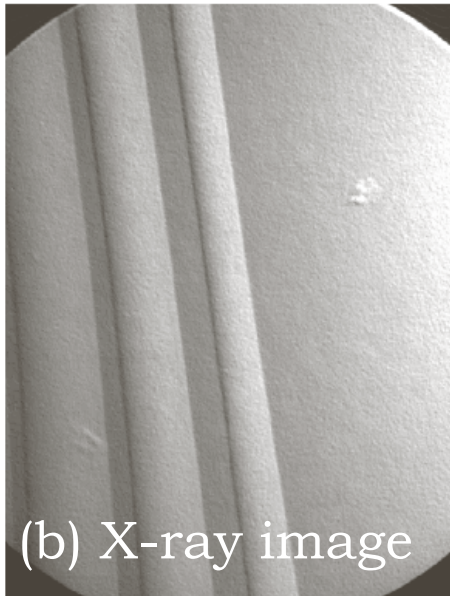
(a)

1	1	1
1	-8	1
1	1	1

a
b c d

FIGURE 10.4

(a) Point detection (Laplacian) mask. (b) X-ray image of turbine blade with a porosity. The porosity contains a single black pixel. (c) Result of convolving the mask with the image. (d) Result of using Eq. (10.2-8) showing a single point (the point was enlarged to make it easier to see). (Original image courtesy of X-TEK Systems, Ltd.)



Point Detection

3. This mask is the Laplacian mask for detecting point. Sum of all weights is zero to make sure that there is no response at a flat region (constant intensity region).
4. The Laplacian is given as (f = input image)

1	1	1
1	-8	1
1	1	1

Laplacian mask

$$\nabla^2 f(x, y) = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2}$$

where

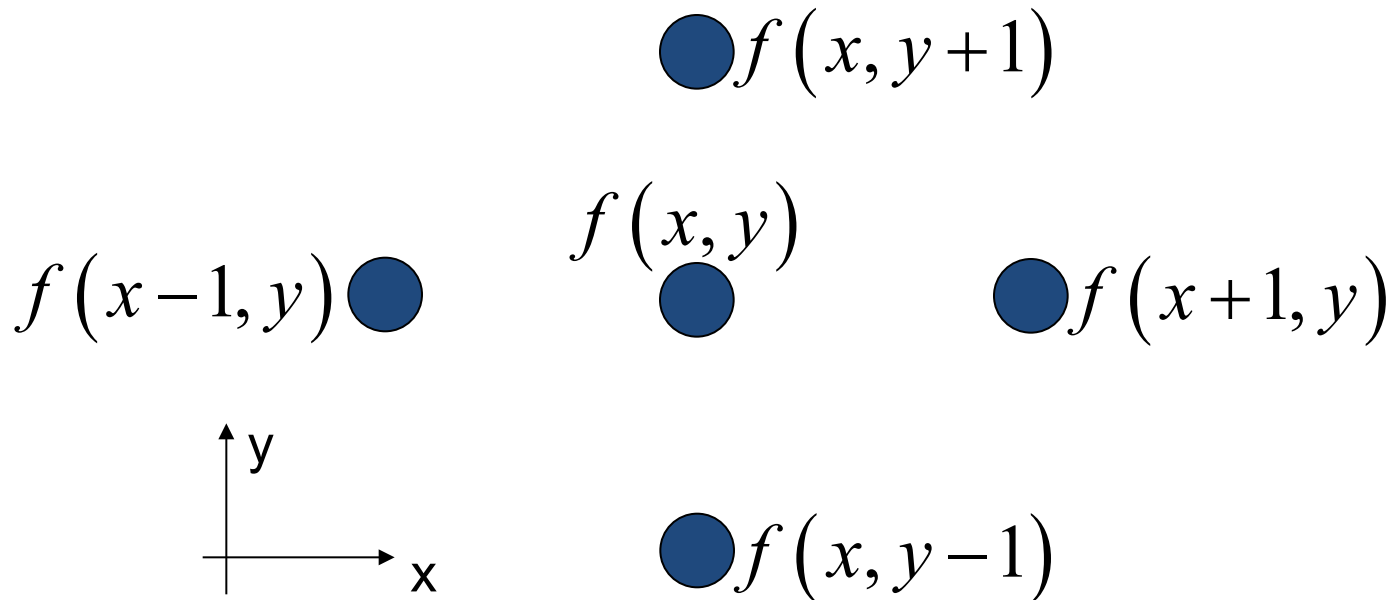
$$\frac{\partial^2 f}{\partial x^2} = f(x+1, y) - 2f(x, y) + f(x-1, y)$$

$$\frac{\partial^2 f}{\partial y^2} = f(x, y+1) - 2f(x, y) + f(x, y-1)$$

Point Detection

5. The discrete implementation of the Laplacian operator is given as (f = input image)

$$\nabla^2 f(x, y) = f(x+1, y) + f(x-1, y) + f(x, y+1) + f(x, y-1) - 4f(x, y)$$



Point Detection

0	1	0	1	1	1
1	-4	1	1	-8	1
0	1	0	1	1	1

0	-1	0	-1	-1	-1
-1	4	-1	-1	8	-1
0	-1	0	-1	-1	-1

a b
c d

FIGURE 3.37

(a) Filter mask used to implement Eq. (3.6-6).

(b) Mask used to implement an extension of this equation that includes the diagonal terms.

(c) and (d) Two other implementations of the Laplacian found frequently in practice.

Line Detection

1. A line is detected when more than one aligned, connected points are detected;
2. Or, the response of line mask is greater than some threshold, e.g.,

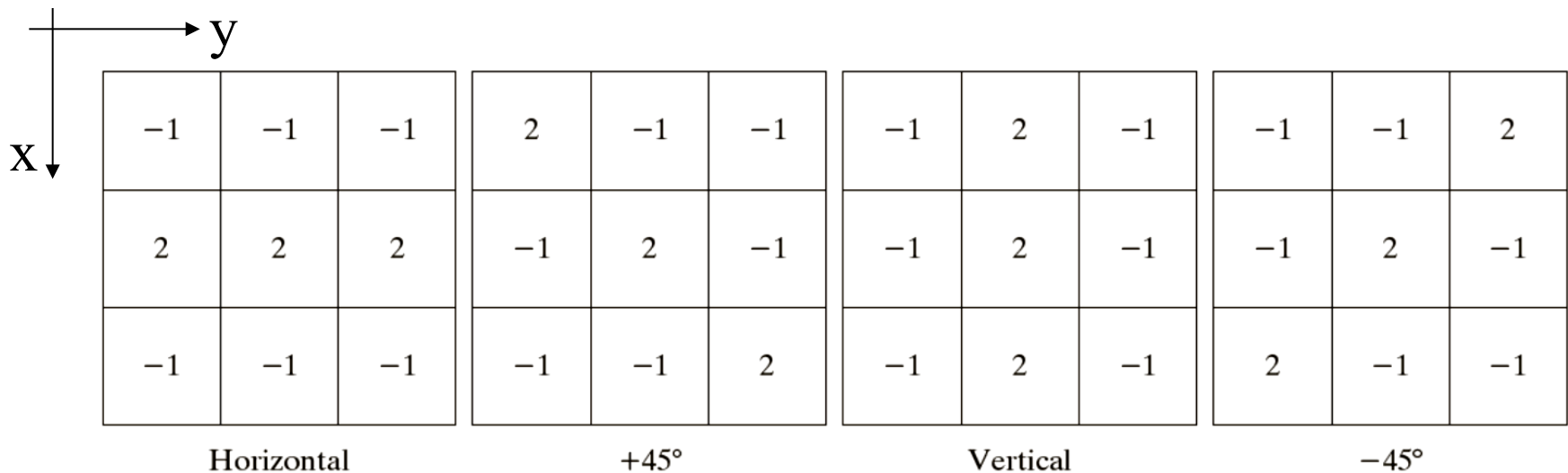


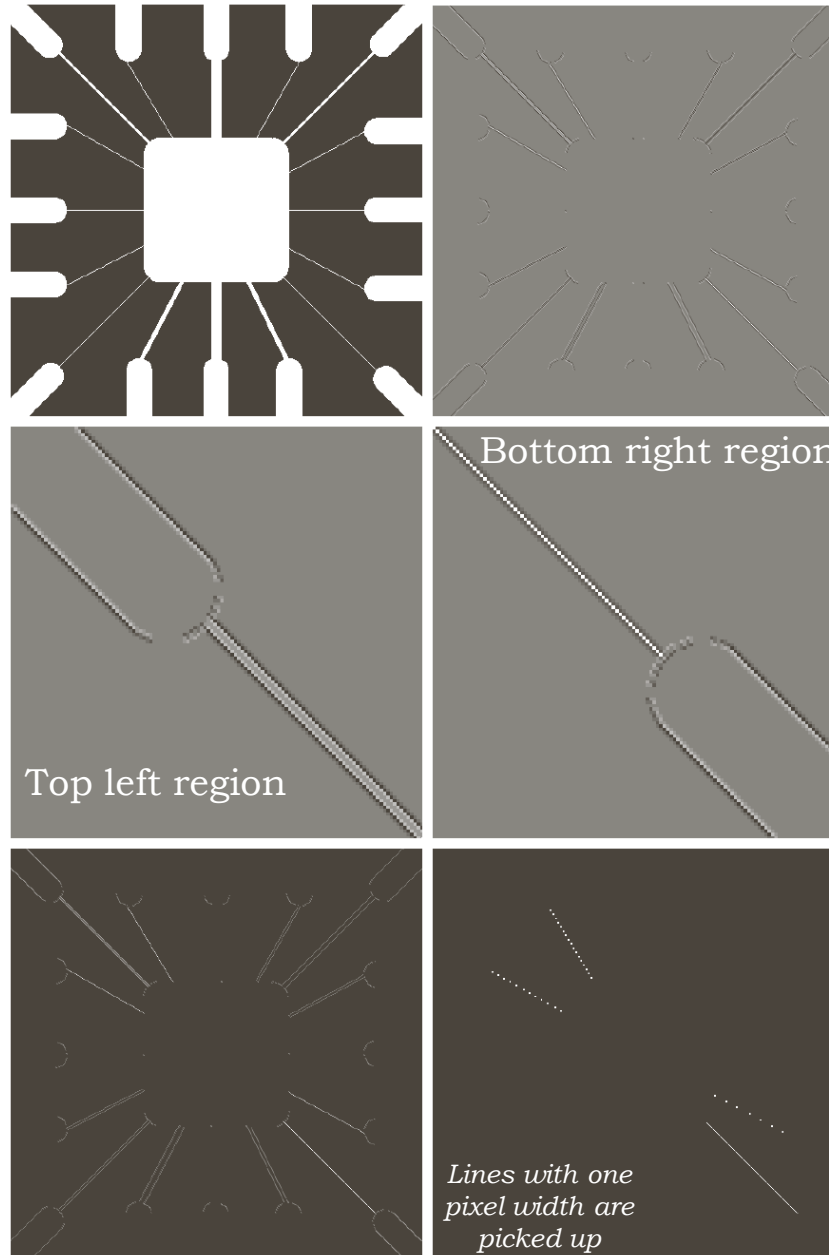
FIGURE 10.6 Line detection masks. Angles are with respect to the axis system in Fig. 2.18(b).

The above are line masks for detecting lines (1 pixel thick) in 4 different specific directions.

Centerline detection: <http://cvlab.epfl.ch/software/centerline-detection>

Line Detection

3. If we want to detect a line in a specified direction, then we should use the mask associated with that direction and threshold its output responses.



a	b
c	d
e	f

FIGURE 10.7
 (a) Image of a wire-bond template.
 (b) Result of processing with the $+45^\circ$ line detector mask in Fig. 10.6.
 (c) Zoomed view of the top left region of (b).
 (d) Zoomed view of the bottom right region of (b).
 (e) The image in (b) with all negative values set to zero.
 (f) All points (in white) whose values satisfied the condition $g \geq T$, where g is the image in (e). (The points in (f) were enlarged to make them easier to see.)

Line Detection

4. If the four line-masks are used, then the final response is the largest response among the masks.

$$R = \max \left(\left| R_{horizontal} \right|, \left| R_{45^\circ} \right|, \left| R_{vertical} \right|, \left| R_{-45^\circ} \right| \right)$$

Edge Detection

1. Edge is the boundary of regions. The boundary has discontinuities in grey intensity level.
2. Three types of edges: ideal edge, ramp edge and roof edge.

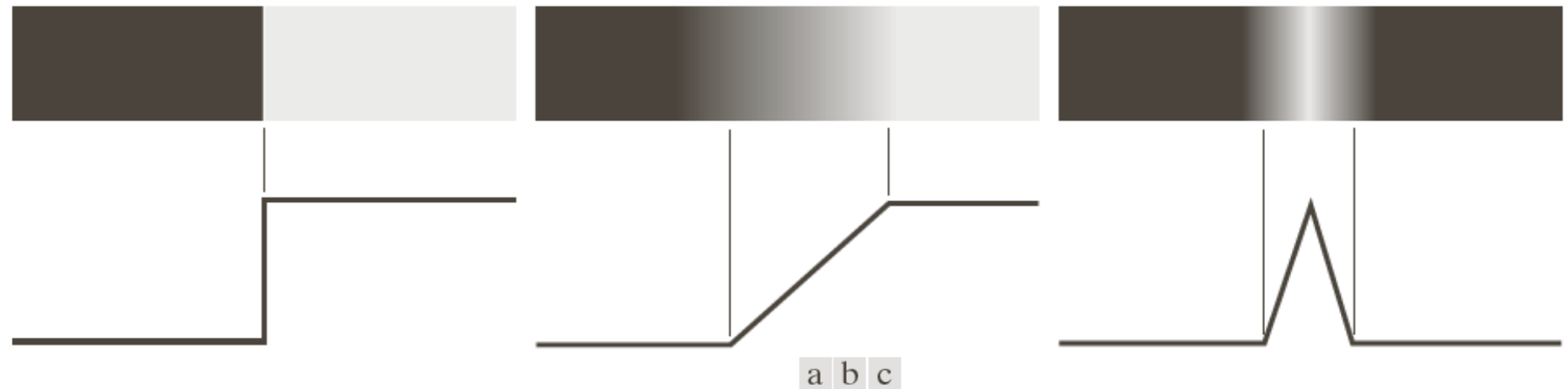


FIGURE 10.8

From left to right, models (ideal representations) of a step, a ramp, and a roof edge, and their corresponding intensity profiles.

Edge Detection

http://en.wikipedia.org/wiki/Edge_detection

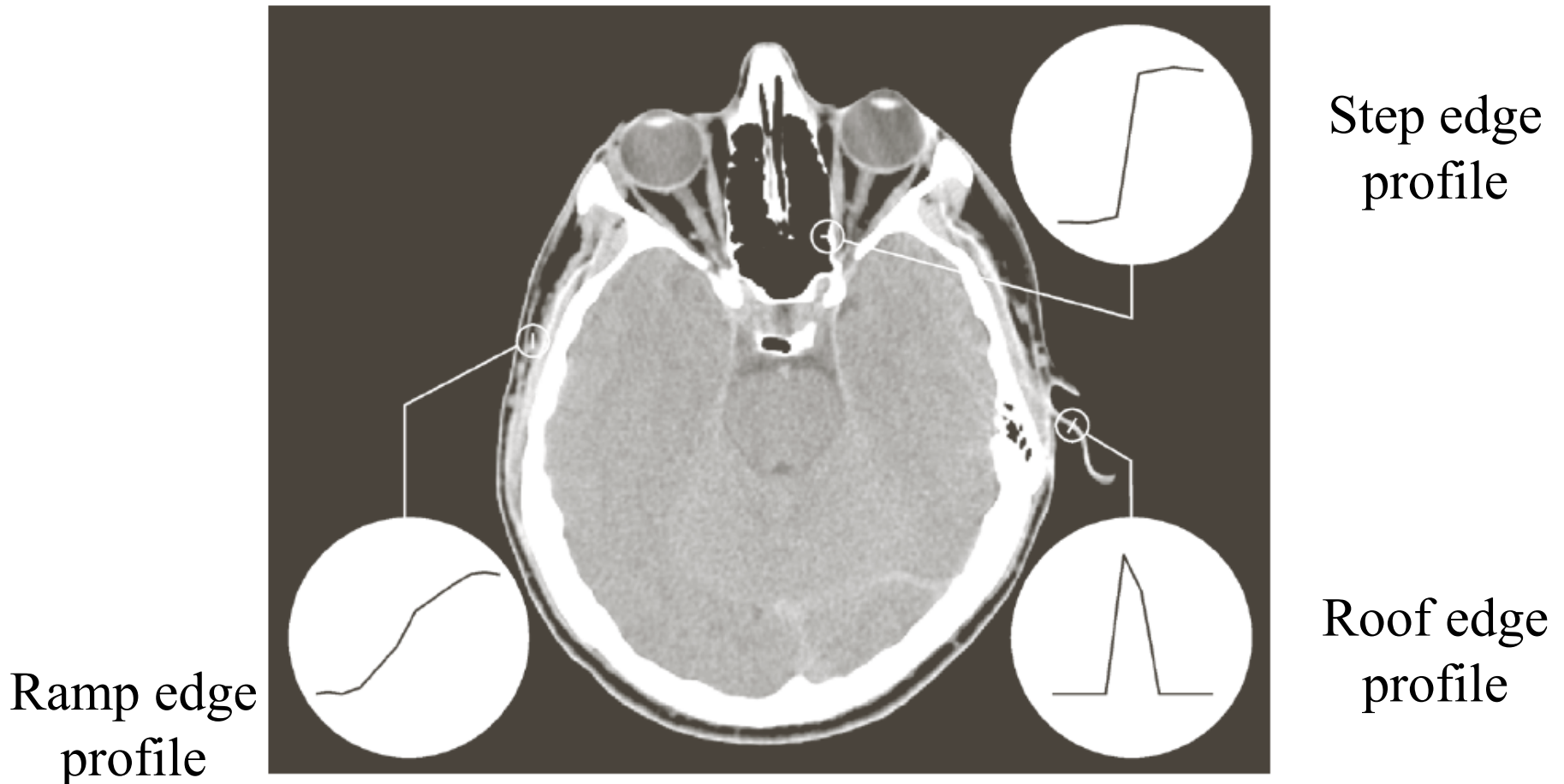


FIGURE 10.9 A 1508×1970 image showing (zoomed) actual ramp (bottom, left), step (top, right), and roof edge profiles. The profiles are from dark to light, in the areas indicated by the short line segments shown in the small circles. The ramp and “step” profiles span 9 pixels and 2 pixels, respectively. The base of the roof edge is 3 pixels. (Original image courtesy of Dr. David R. Pickens, Vanderbilt University.)

The figure is provided by Pearson Education, Digital Image Processing, Gonzalez & Woods, www.ImageProcessingPlace.com

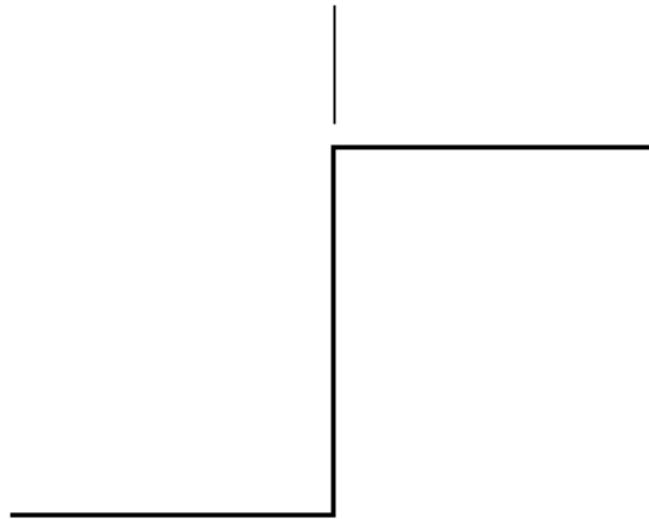
Edge Detection

3. For an ideal edge (step edge), an edge is a collection of connected pixels on the region boundary. Ideal edges can occur over the distance of 1 pixel.

Model of an ideal digital edge



Gray-level profile
of a horizontal line
through the image



Edge Detection

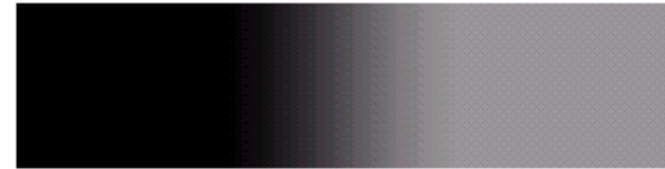
4. Roof edges are models of lines through a region, with the base (width) of a roof edge being determined by the thickness and sharpness of the line.
5. In the limit, when its base is 1 pixel wide, a roof edge becomes a 1 pixel thick line running through a region in an image.
6. Roof edges can represent thin features, e.g., roads, line drawings, etc.



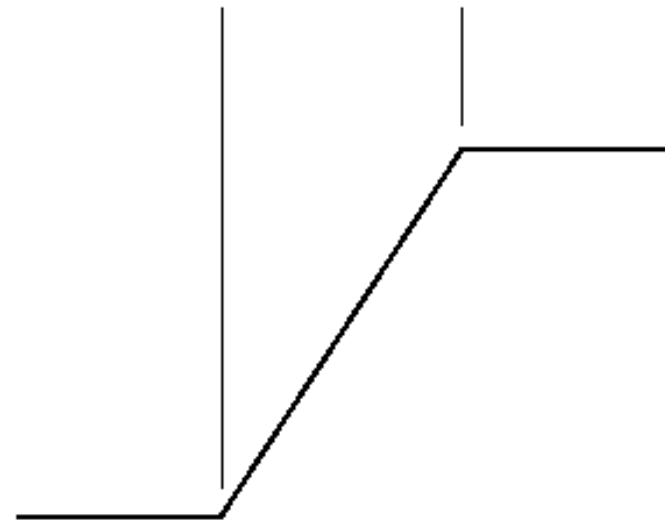
Edge Detection

7. For a ramp edge,
- a. edge point is any point contained in the ramp.
 - b. edge length is determined by the length of the ramp.
 - c. the slope of the ramp is inversely proportional to the degree of blurring in the edge.
 - d. the first derivative of the intensity profile is positive at the points of transition into and out of the ramp (we move from left to right).
 - e. the second derivative of the intensity profile is positive at the transition associated with the dark side of the edge, and negative at the transition associated with the light side of the edge.

Model of a ramp digital edge



Gray-level profile of a horizontal line through the image

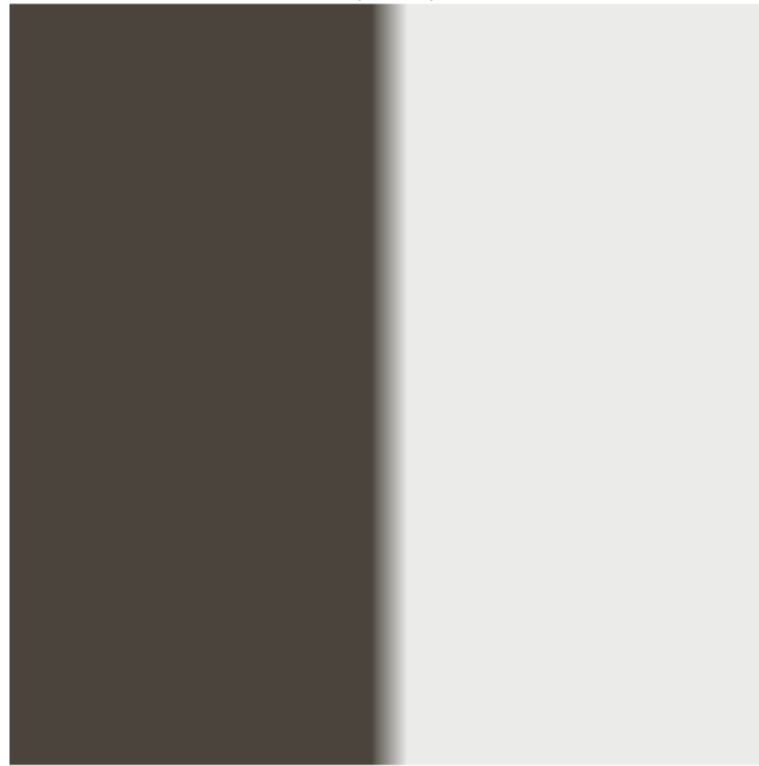


Edge Detection

a b

FIGURE 10.10

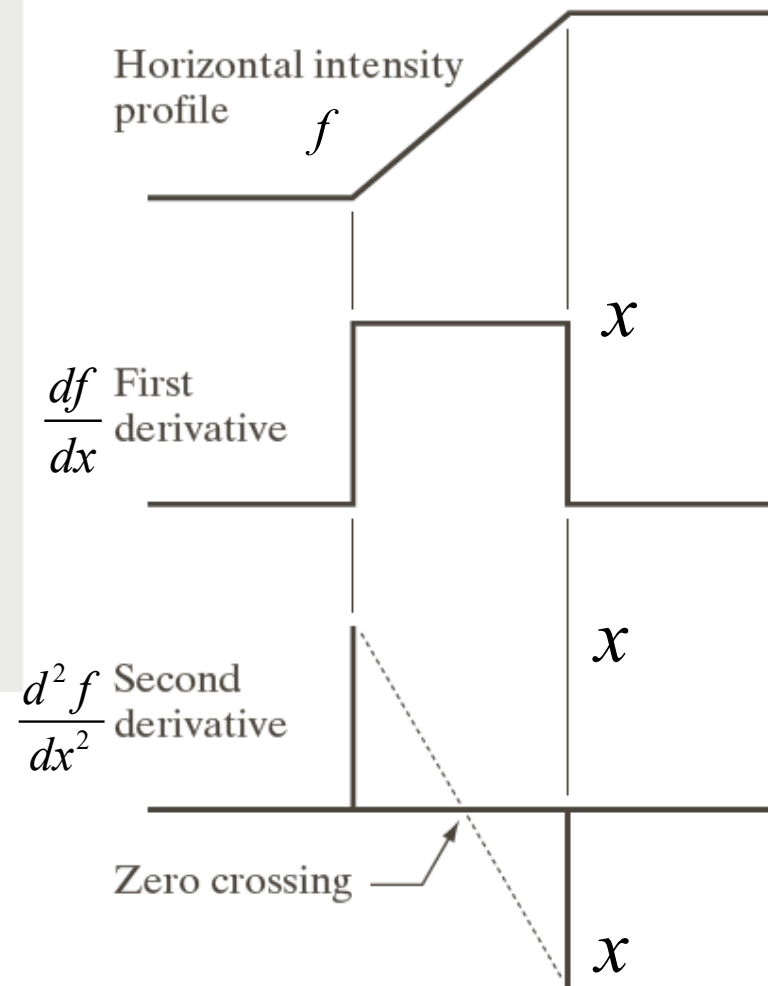
(a) Two regions of constant intensity separated by an ideal vertical ramp edge.
(b) Detail near the edge, showing a horizontal intensity profile, together with its first and second derivatives.



Finite difference approximations

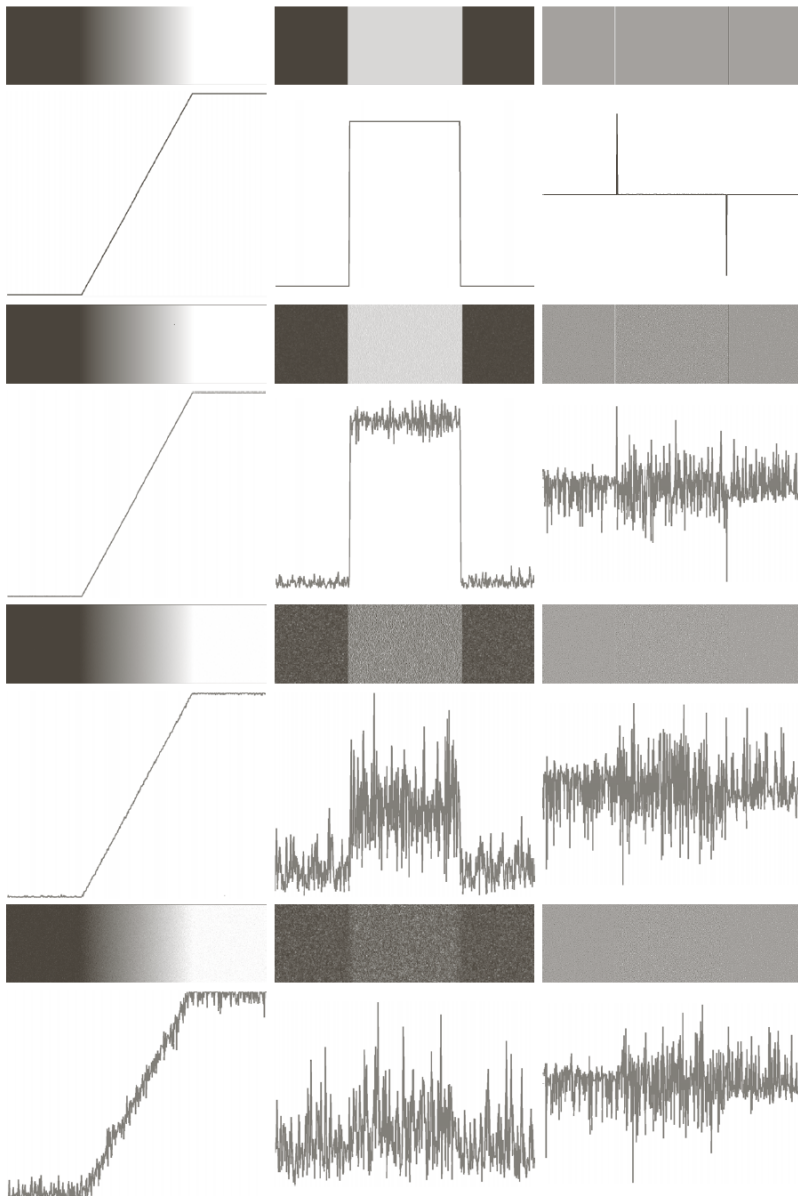
$$\frac{df}{dx} \approx f[x+1] - f[x]$$

$$\frac{d^2f}{dx^2} \approx f[x+1] - 2f[x] + f[x-1]$$



Edge Detection

8. The magnitude of the first derivative can be used to detect the presence of an edge.
9. The sign of the second derivative can be used to determine whether an edge pixel lies on the dark or light side of an edge.
10. The *zero-crossing* property of the second derivative is very useful for locating the centres of thick edges.
11. However, fairly little noise can have a significant impact on the first and second derivatives used for edge detection in images.
12. Image smoothing is commonly used prior to edge detection to improve the accuracy of the estimates of the two derivatives.



1. Noise free image.
2. The first and second derivatives are estimated correctly.

1. Noisy image.
2. The estimations of the first and second derivatives are corrupted.

FIGURE 10.11 First column: Images and intensity profiles of a ramp edge corrupted by random Gaussian noise of zero mean and standard deviations of 0.0, 0.1, 1.0, and 10.0 intensity levels, respectively. Second column: First-derivative images and intensity profiles. Third column: Second-derivative images and intensity profiles.

Gradient operator

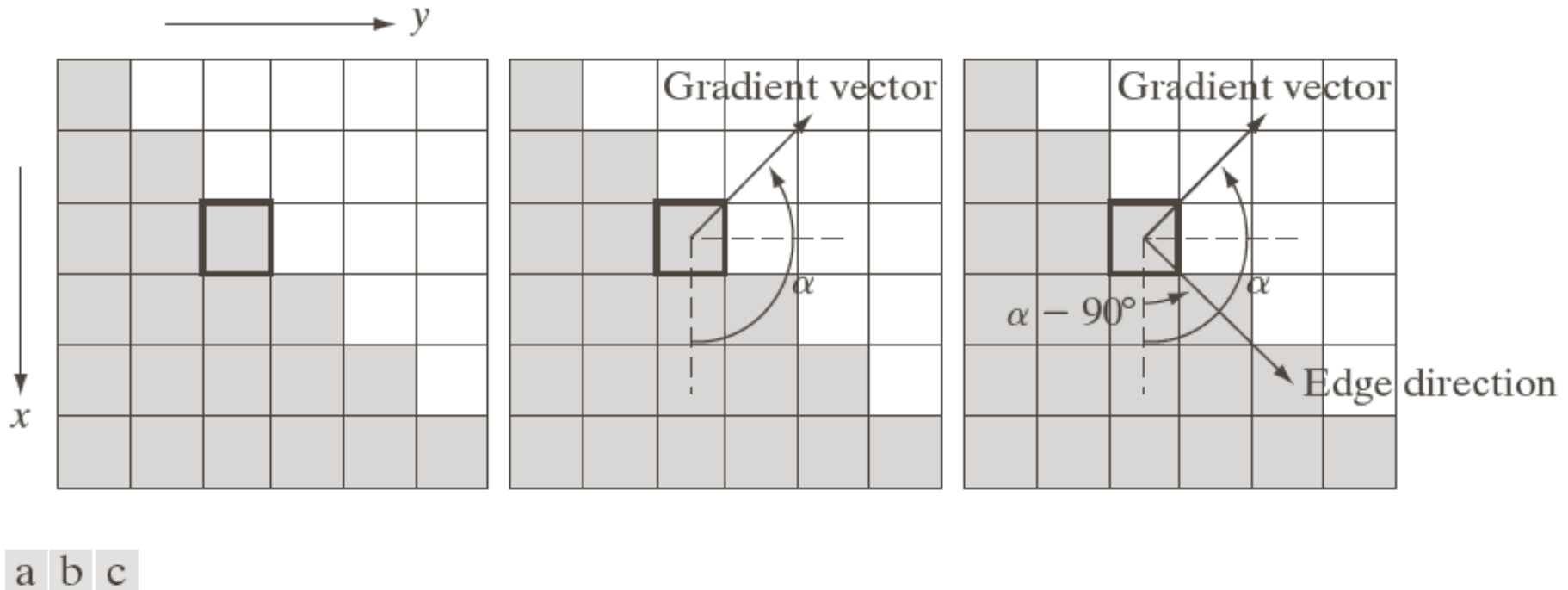


FIGURE 10.12 Using the gradient to determine edge strength and direction at a point. Note that the edge is perpendicular to the direction of the gradient vector at the point where the gradient is computed. Each square in the figure represents one pixel.

Gradient operator

13. The gradient of an image $f(x,y)$ at location (x,y) is defined as

$$\nabla f = \text{grad}(f) = \begin{bmatrix} G_x \\ G_y \end{bmatrix} = \begin{bmatrix} \frac{\partial f}{\partial x} \\ \frac{\partial f}{\partial y} \end{bmatrix} \quad \text{where} \quad \begin{aligned} \frac{\partial f}{\partial x} &\approx f[x+1, y] - f[x, y] \\ \frac{\partial f}{\partial y} &\approx f[x, y+1] - f[x, y] \end{aligned}$$

14. Gradient Magnitude

$$|\nabla f| = \text{mag}(\nabla f) = \sqrt{\left(\frac{\partial f}{\partial x}\right)^2 + \left(\frac{\partial f}{\partial y}\right)^2}$$

Gradient operator

15. Gradient direction

$$\tan^{-1} \left(\frac{G_y}{G_x} \right)$$

16. The computation of the gradient of an image is based on obtaining the partial derivatives $\partial f / \partial x$ and $\partial f / \partial y$ at every pixel location (x,y) .

a	
b	c
d	e
f	g

FIGURE 10.14

A 3×3 region of an image (the z 's are intensity values) and various masks used to compute the gradient at the point labeled z_5 .

For horizontal
edge detection

For horizontal
edge detection

z_1	z_2	z_3
z_4	z_5	z_6
z_7	z_8	z_9

where z_i represent
intensity values.

-1	0	0	-1
0	1	1	0

Roberts

-1	-1	-1	-1	0	1
0	0	0	-1	0	1
1	1	1	-1	0	1

For vertical
edge detection

Prewitt

-1	-2	-1	-1	0	1
0	0	0	-2	0	2
1	2	1	-1	0	1

For vertical
edge detection

Sobel

Gradient operator

17. Roberts cross-gradient operators

$$G_x = (z_9 - z_5) \quad \text{where } z_i \text{ represent intensity values.}$$
$$G_y = (z_8 - z_6)$$

z_1	z_2	z_3
z_4	z_5	z_6
z_7	z_8	z_9

18. Prewitt operators

$$G_x = (z_7 + z_8 + z_9) - (z_1 + z_2 + z_3) \quad \text{where } z_i \text{ represent intensity values.}$$
$$G_y = (z_3 + z_6 + z_9) - (z_1 + z_4 + z_7)$$

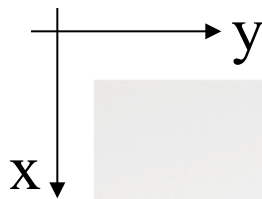
19. Sobel operators

$$G_x = (z_7 + 2z_8 + z_9) - (z_1 + 2z_2 + z_3) \quad \text{where } z_i \text{ represent intensity values.}$$
$$G_y = (z_3 + 2z_6 + z_9) - (z_1 + 2z_4 + z_7)$$

Gradient operator

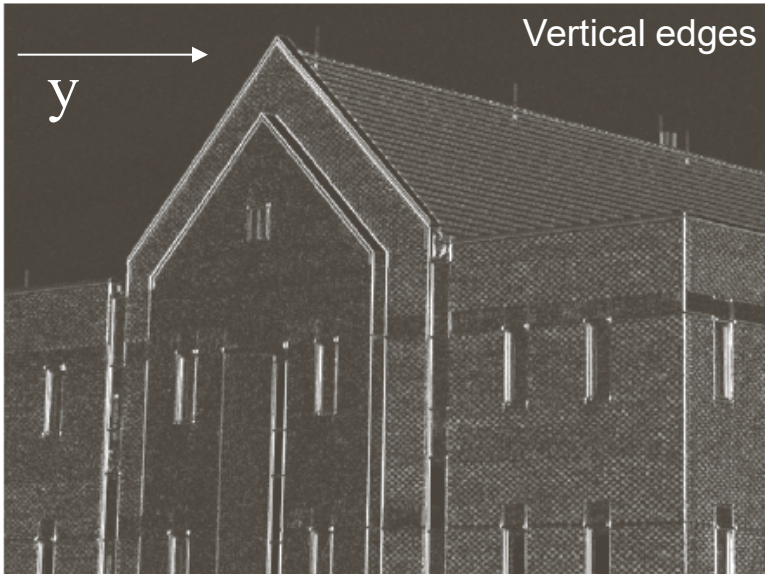
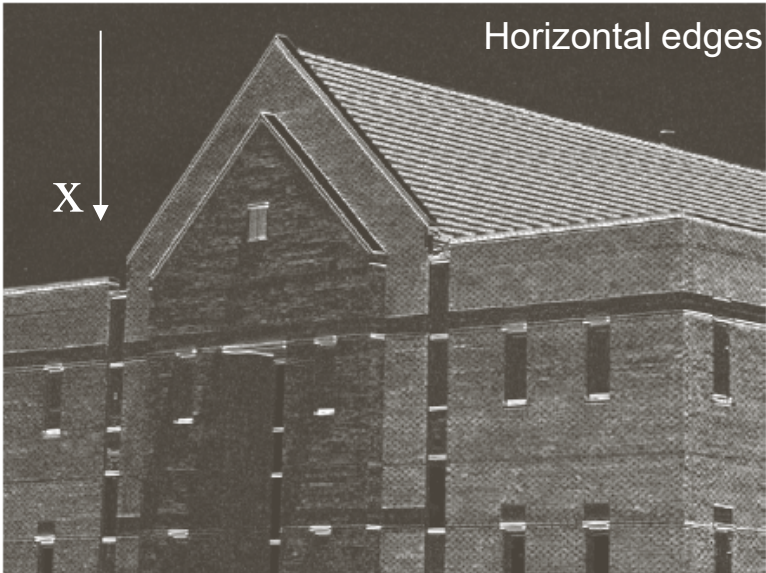
20. Approximation

$$|\nabla f| \approx |G_x| + |G_y|$$



a	b
c	d

FIGURE 10.16
 (a) Original image of size 834×1114 pixels, with intensity values scaled to the range $[0, 1]$.
 (b) $|g_x|$, the component of the gradient in the x -direction, obtained using the Sobel mask in Fig. 10.14(f) to filter the image.
 (c) $|g_y|$, obtained using the mask in Fig. 10.14(g).
 (d) The gradient image, $|g_x| + |g_y|$.





a	b
c	d

FIGURE 10.18
 Same sequence as in Fig. 10.16, but with the original image smoothed using a 5×5 averaging filter prior to edge detection.



Gradient operator

23. Diagonal edge masks for detecting discontinuities in the diagonal directions.

0	1	1
-1	0	1
-1	-1	0

Prewitt

0	1	2
-1	0	1
-2	-1	0

Sobel

-1	-1	0
-1	0	1
0	1	1

z_1	z_2	z_3
z_4	z_5	z_6
z_7	z_8	z_9

a	b
c	d

FIGURE 10.15
Prewitt and Sobel
masks for
detecting diagonal
edges.



a b



FIGURE 10.19

Diagonal edge
detection.

(a) Result of
using the mask in
Fig. 10.15(c).

(b) Result of
using the mask in
Fig. 10.15(d). The
input image in
both cases was
Fig. 10.18(a).

Combining the gradient with thresholding



a b

FIGURE 10.20 (a) Thresholded version of the image in Fig. 10.16(d), with the threshold selected as 33% of the highest value in the image; this threshold was just high enough to eliminate most of the brick edges in the gradient image. (b) Thresholded version of the image in Fig. 10.18(d), obtained using a threshold equal to 33% of the highest value in that image.

The Laplacian

1. The Laplacian of an image $f(x,y)$ at location (x,y) is defined as

$$\nabla^2 f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2}$$

2. Two approximations

$$\nabla^2 f = 4z_5 - (z_2 + z_4 + z_6 + z_8)$$

$$\nabla^2 f = 8z_5 - (z_1 + z_2 + z_3 + z_4 + z_6 + z_7 + z_8 + z_9)$$

The Laplacian

3. Two masks for the approximations

0	-1	0	-1	-1	-1
-1	4	-1	-1	8	-1
0	-1	0	-1	-1	-1

4. The Laplacian generally is not used in its original form for edge detection (based on the zero-crossing property) because it is unacceptably sensitive to noise.
5. Solution: smooth the image by using a Gaussian blurring function $h(r)$ before we apply the Laplacian operator.

$$h(r) = -e^{\frac{-r^2}{2\sigma^2}}$$

The Laplacian

6. *The Laplacian of a Gaussian (LoG) operator*

$$LoG(f) = \nabla^2 (f * h)$$

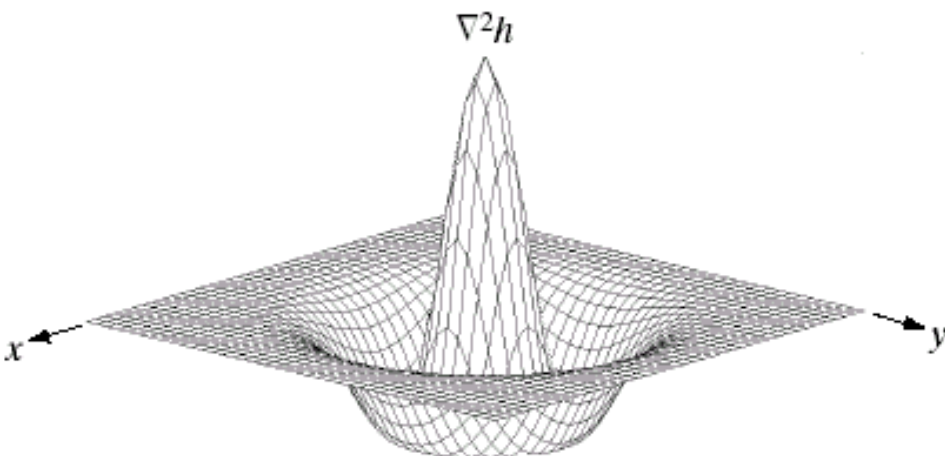
7. Another definition

$$LoG(f) = f * (\nabla^2 h)$$

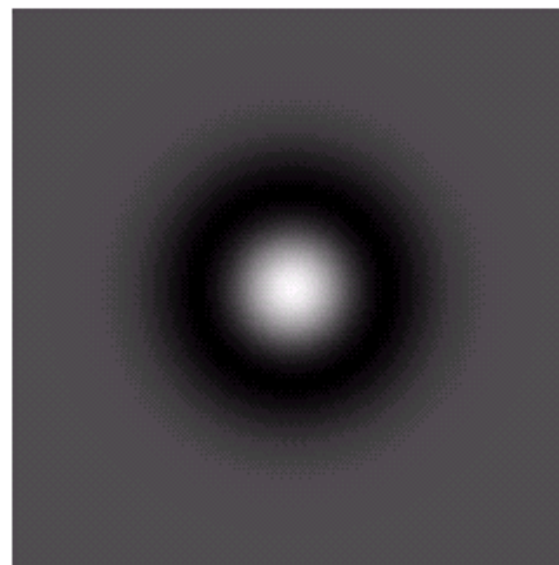
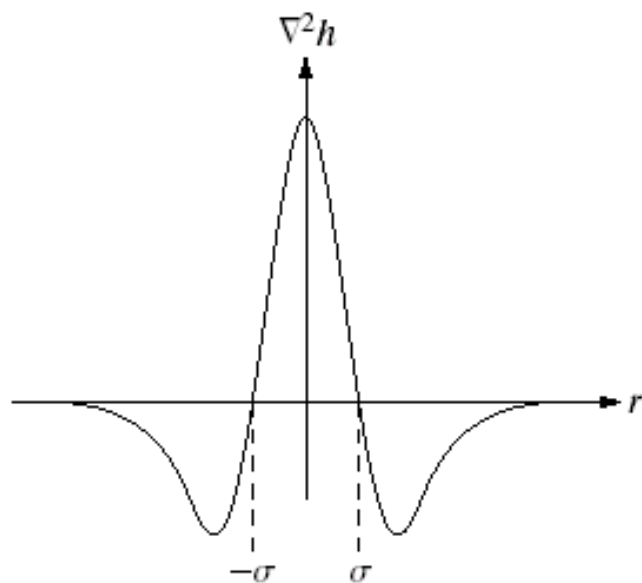
where

$$\nabla^2 h = - \left[\frac{r^2 - 2\sigma^2}{\sigma^4} \right] e^{\frac{-r^2}{2\sigma^2}}$$

8. Using the LoG, the location of edges can be detected reliably based on the zero-crossing values. It is because noise in the image is reduced by the Gaussian function.



Mexican hat function



a b
c d

FIGURE 10.14

Laplacian of a Gaussian (LoG).

(a) 3-D plot.

(b) Image (black is negative, gray is the zero plane, and white is positive).

(c) Cross section showing zero crossings.

(d) 5×5 mask approximation to the shape of (a).

0	0	-1	0	0
0	-1	-2	-1	0
-1	-2	16	-2	-1
0	-1	-2	-1	0
0	0	-1	0	0

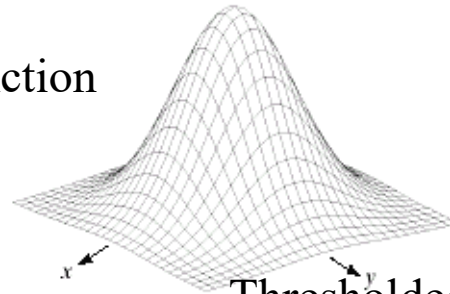
Original image



Sobel gradient
magnitudes



Gaussian function



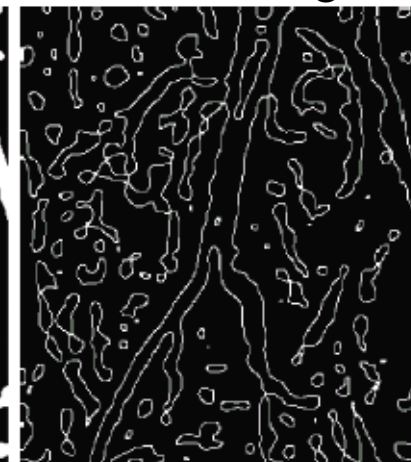
-1	-1	-1
-1	8	-1
-1	-1	-1

Laplacian mask

Thresholded LoG

Zero-crossings

Laplacian of
Gaussian (LoG)
filter responses



a b
c d
e f g

FIGURE 10.15 (a) Original image. (b) Sobel gradient (shown for comparison). (c) Spatial Gaussian smoothing function. (d) Laplacian mask. (e) LoG. (f) Thresholded LoG. (g) Zero crossings. (Original image courtesy of Dr. David R. Pickens, Department of Radiology and Radiological Sciences, Vanderbilt University Medical Center.)

The Laplacian in frequency domain

1. The Laplacian of an image $f(x,y)$ at location (x,y) is defined as

$$\nabla^2 f(x,y) = \frac{\partial^2 f(x,y)}{\partial x^2} + \frac{\partial^2 f(x,y)}{\partial y^2}$$

2. Frequency representation

$$\mathfrak{F}\{\nabla^2 f(x,y)\} = -(u^2 + v^2)F(u,v)$$

3. LoG in frequency domain?

Canny edge detector

- The Canny edge detector is one of the most popular detectors of edge pixels. http://en.wikipedia.org/wiki/Canny_edge_detector
- The edge detection process serves to simplify the analysis of images by drastically reducing the amount of data to be processed, while at the same time preserving useful structural information about object boundaries.
- There are two common criteria relevant to edge detector performance.
 - It is important that edges that occur in the image should not be missed and that there be no spurious responses.
 - The edge points should be well localized. That is, the distance between the points marked by the detector and the true edge center should be minimized.
- One additional requirement is to circumvent the possibility of multiple responses to a single edge.

Canny edge detector

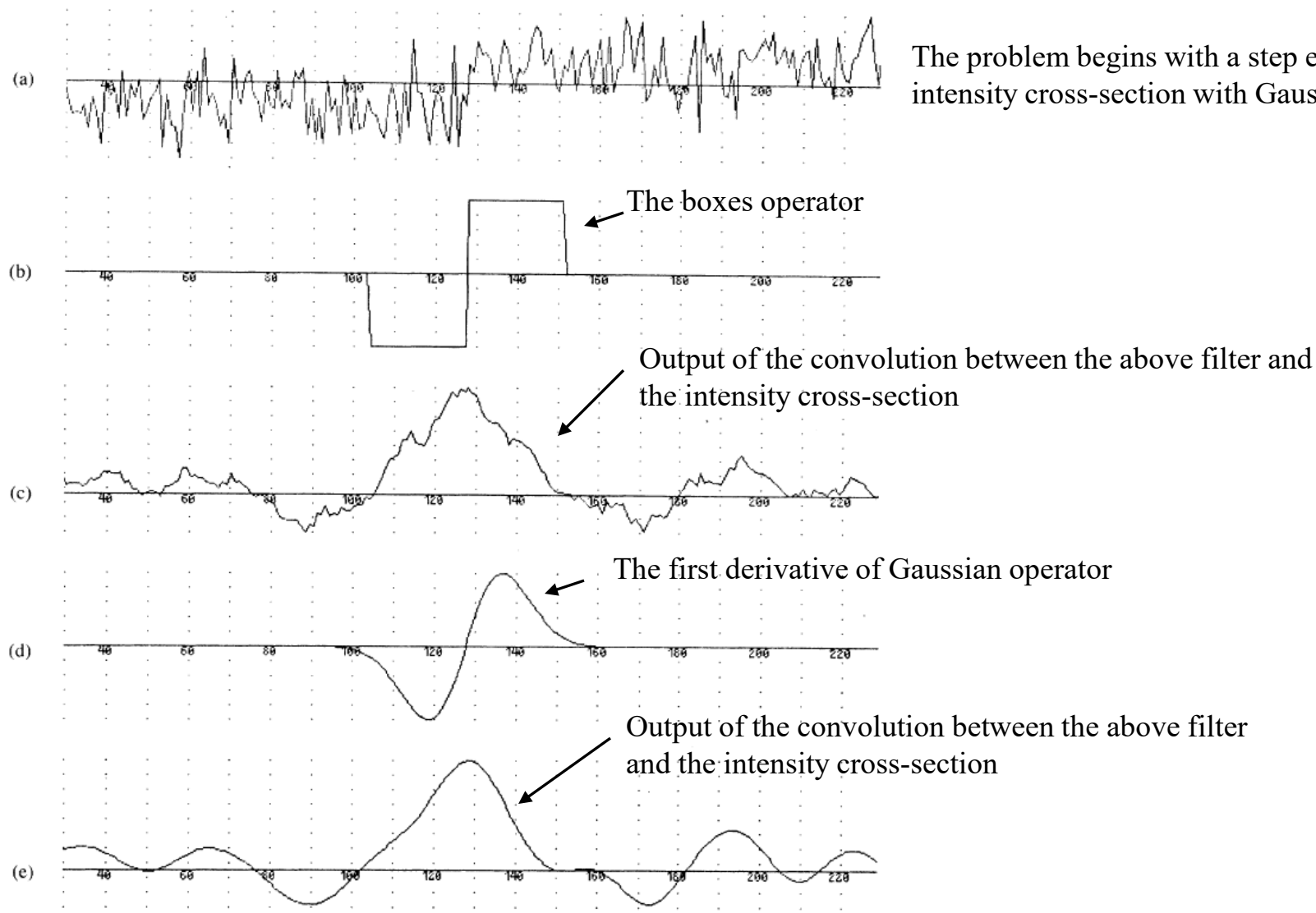
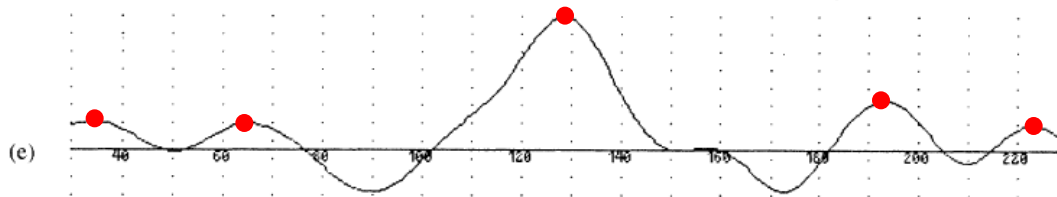
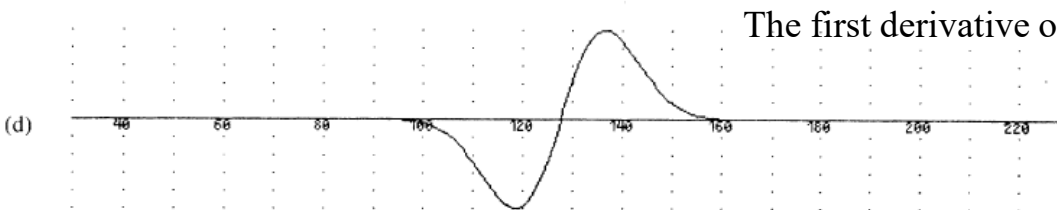
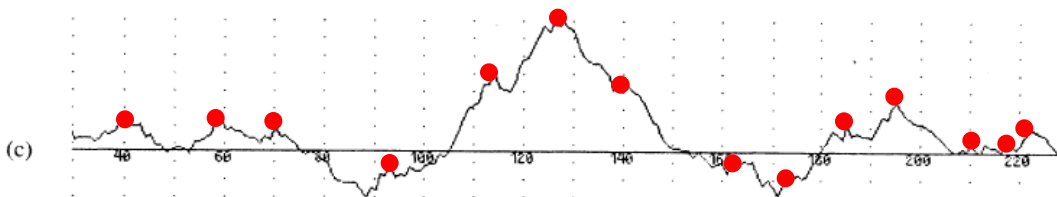
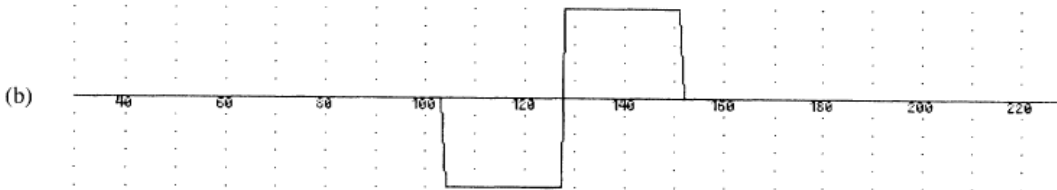
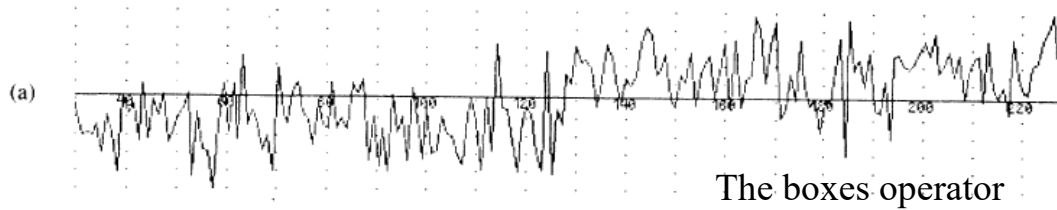


Fig. 1. (a) A noisy step edge. (b) Difference of boxes operator. (c) Difference of boxes operator applied to the edge. (d) First derivative of Gaussian operator. (e) First derivative of Gaussian applied to the edge.

Canny edge detector



- (1) The edge center is marked as red circle at a local maximum in the output of the filter responses.
- (2) Within the region of the edge, the boxes operator exhibits more local maxima than the Gaussian operator.
- (3) Therefore, the Gaussian operator is better than the boxes operator in this example.

Fig. 1. (a) A noisy step edge. (b) Difference of boxes operator. (c) Difference of boxes operator applied to the edge. (d) First derivative of Gaussian operator. (e) First derivative of Gaussian applied to the edge.

The three performance criteria

- [1] Good detection (low error rate) → Detection Criterion

- There should be a low probability of failing to mark real edge points, and a low probability of falsely marking non-edge points.

We define our first criterion, the output signal-to-noise ratio, as the quotient of these two responses.

$$\text{SNR} = \frac{\left| \int_{-W}^{+W} G(-x) f(x) dx \right|}{n_0 \sqrt{\int_{-W}^{+W} f^2(x) dx}} \quad (3)$$

- [2] Good localization (edge points should be well localized) → Localization Criterion

- The points marked as edge points by the operator should be as close as possible to the true edge center.

The localization is defined as the reciprocal of δx_0 .

$$\text{Localization} = \frac{\left| \int_{-W}^{+W} G'(-x) f'(x) dx \right|}{n_0 \sqrt{\int_{-W}^{+W} f'^2(x) dx}} \quad (9)$$

- [3] Only one response to a single edge (single edge point response) → Multiple Response Constraint

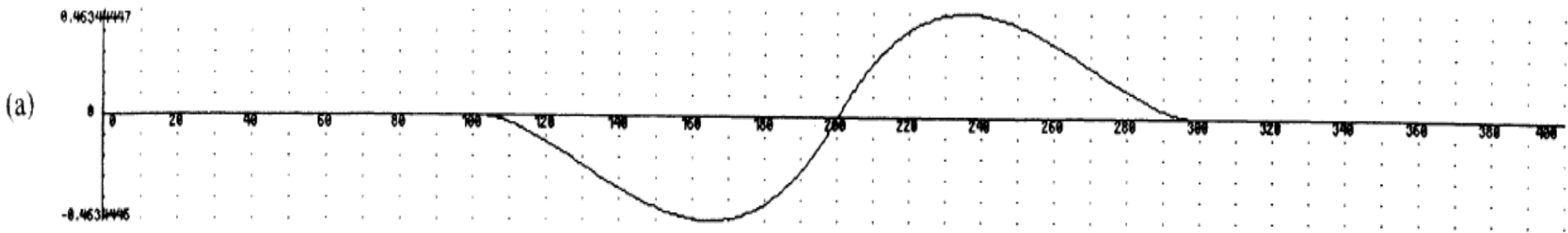
$$x_{zc}(f) = \pi \left(\frac{\int_{-\infty}^{+\infty} f'^2(x) dx}{\int_{-\infty}^{+\infty} f''^2(x) dx} \right)^{1/2} \quad (12)$$

Finding optimal detectors by numerical optimization

- It is very difficult to find the function f (filter) which maximizes the detection and localization criteria subject to the multiple response constraint. Numerical optimization is therefore used.
- The optimal solution of the previous formulations is of the form.

$$f(x) = a_1 e^{\alpha x} \sin \omega x + a_2 e^{\alpha x} \cos \omega x + a_3 e^{-\alpha x} \cdot \sin \omega x + a_4 e^{-\alpha x} \cos \omega x + c. \quad (35)$$

- The variables are determined by the non-linear optimization with boundary conditions.



Optimal detector approximation

- The numerically estimated optimal edge detector can be approximated by the first derivative of a Gaussian G .

$$G(x) = \exp\left(-\frac{x^2}{2\sigma^2}\right) \quad f(x) = G'(x) = -\frac{x}{\sigma^2} e^{\frac{-x^2}{2\sigma^2}}$$

- For 2D, the solution proposed by Canny amounts to convolving the initial image with a Gaussian function, followed by the computation of the derivatives in x and y of the result.

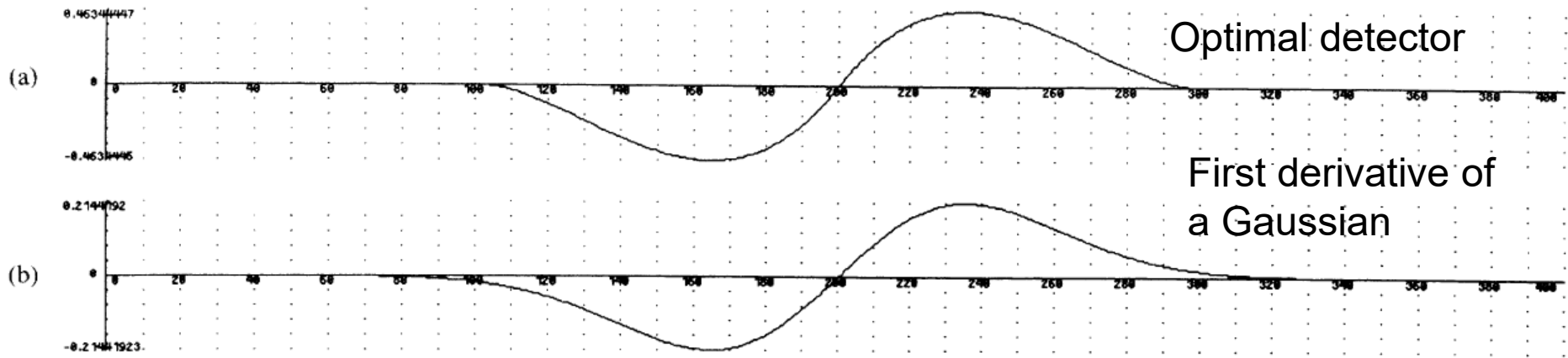


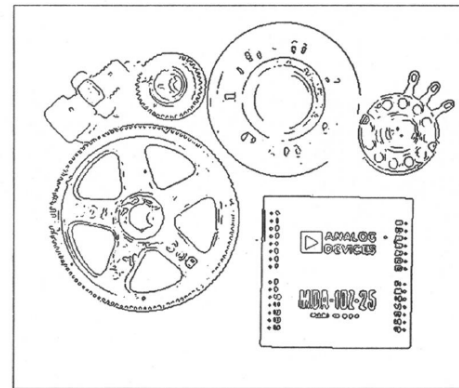
Fig. 6. (a) The optimal step edge operator. (b) The first derivative of a Gaussian.

Canny edge detection algorithm

- There are three main steps in the Canny edge detection
- [1] Gradient calculation
- [2] Non-maximum suppression
- [3] Hysteresis thresholding



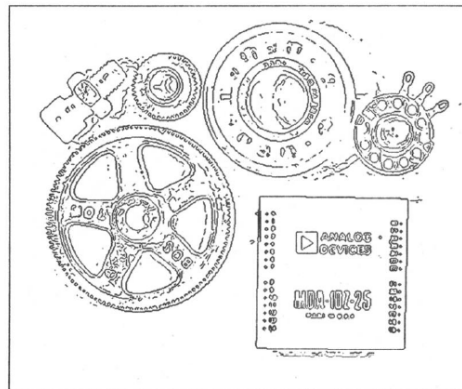
(a)



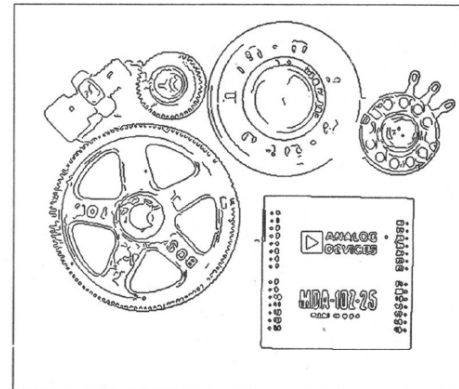
(c)

T_{high}

T_{low}



(b)



(d)

Hysteresis
thresholding

Non-maximum suppression

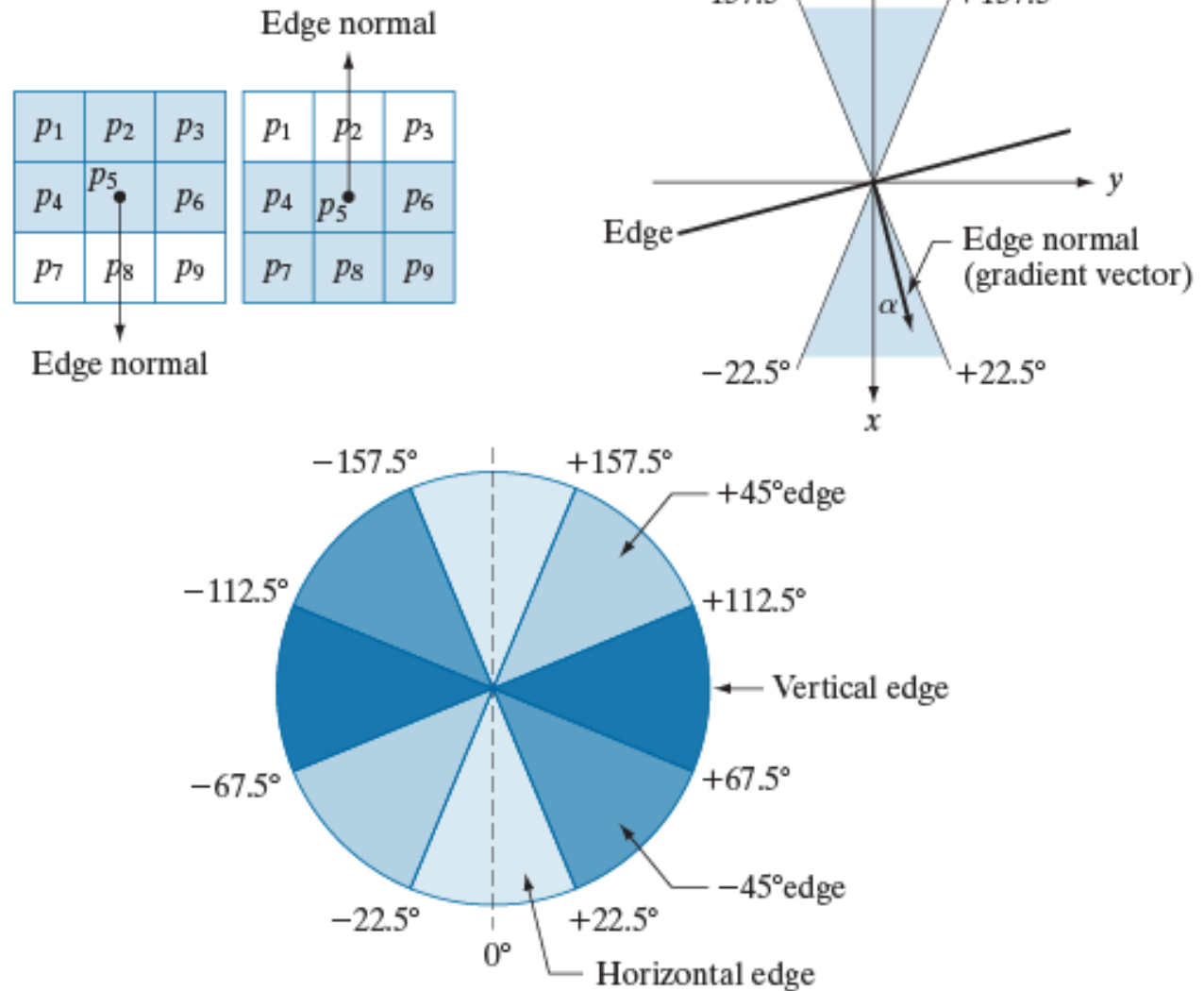
- Non-maximum suppression is to thin the wide ridges around local maxima.
- Four discrete orientations of the edge normal (gradient) are specified: horizontal edge, -45-degree edge, vertical edge, +45-degree edge.
- The non-maximum suppression scheme for a 3x3 region is the following.
 1. Find the edge normal direction d_k that is closest to $\alpha(x, y)$ at a point (x, y)
 2. Let K denote the value of $|\nabla f_s(x, y)|$ at (x, y) , where f_s represents the smoothed image. If K is less than the value of $|\nabla f_s|$ at one or both of the neighbors of point (x, y) along d_k , then $g_N(x, y) = 0$ (suppression is performed); otherwise, set $g_N(x, y) = K$.

Non-maximum suppression

a b
c

FIGURE 10.24

(a) Two possible orientations of a horizontal edge (shaded) in a 3×3 neighborhood. (b) Range of values (shaded) of α , the direction angle of the edge normal for a horizontal edge. (c) The angle ranges of the edge normals for the four types of edge directions in a 3×3 neighborhood. Each edge direction has two ranges, shown in corresponding shades.



Hysteresis thresholding

- The hysteresis thresholding is to reduce false edge points by using two thresholds: a low threshold T_L and a high threshold T_H .
- If we set the threshold too low, there will be some false edges (false positives). If we set the threshold too high, we will get some false negatives (valid edge points will be eliminated).
- We form two images, $g_{NH}(x, y)$ and $g_{NL}(x, y)$
 - $g_{NH}(x, y) = g_N(x, y) \geq T_H$
 - $g_{NL}(x, y) = g_N(x, y) \geq T_L$
 - $g_{NL}(x, y) = g_{NL}(x, y) - g_{NH}(x, y)$
- The nonzero pixels in g_{NH} and g_{NL} may be viewed as being “strong” and “weak” edge pixels, respectively.

Hysteresis thresholding

- All nonzero pixels in g_{NH} are marked as valid edge pixels and marked unvisited.
- The following procedure is implemented to expand the edge regions.
 1. Locate the next unvisited edge pixel, p , in $g_{NH}(x, y)$, and mark it visited.
 2. Mark as valid edge pixels all the weak pixels in $g_{NL}(x, y)$ that are connected to p using, say, 8-connectivity.
 3. If all nonzero pixels in $g_{NH}(x, y)$ have been visited go to Step (4). Else, return to Step (1).
 4. Set to zero all pixels in $g_{NL}(x, y)$ that were not marked as valid edge pixels.
- At the end of this procedure, the final image output by the above algorithm is formed by appending to $g_{NH}(x, y)$ all the nonzero pixels from $g_{NL}(x, y)$.

Hysteresis thresholding

- The two images g_{NH} and g_{NL} are used for discussion purpose. In practice, the algorithm is implemented directly on g_N by forming a list of strong pixels and the weak pixels connected to them.

a	b
c	d

FIGURE 10.26

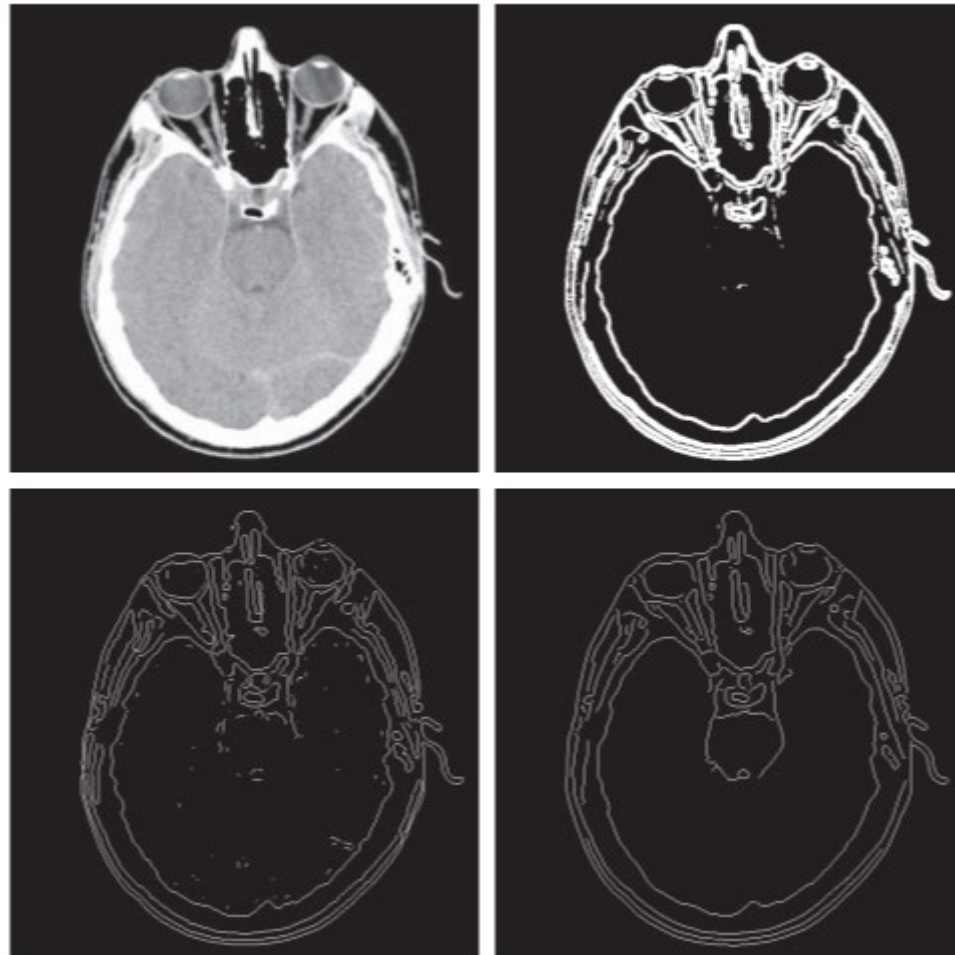
(a) Head CT image of size 512×512 pixels, with intensity values scaled to the range $[0, 1]$.

(b) Thresholded gradient of the smoothed image.

(c) Image obtained using the Marr-Hildreth algorithm.

(d) Image obtained using the Canny algorithm.

(Original image courtesy of Dr. David R. Pickens, Vanderbilt University.)

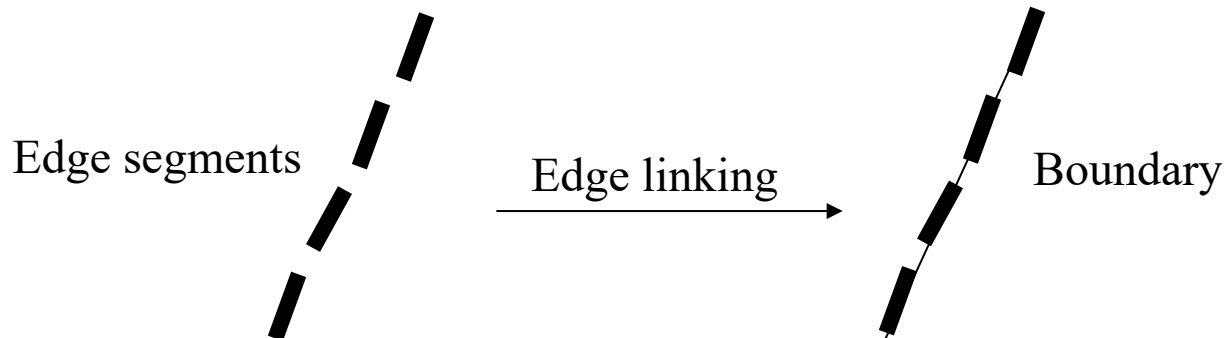


Hysteresis thresholding

- Discussion (in class)

Edge linking and boundary detection

1. Edge points/pixels seldom characterise an edge completely.
2. Reason: noise, breaks in the edge from non-uniform illumination, etc.
3. Solutions: edge points are linked to become a meaningful boundary by using either
 - a. Local processing, or
 - b. Global processing (based on Hough Transform).



Local processing

1. Idea: all edge points that are similar according to a set of predefined criteria are linked/connected.
2. Criterion 1: similar strength of the response of the gradient operator, which is used to produce an edge pixel.

$$\left| \left| \nabla f(x, y) \right| - \left| \nabla f(x_0, y_0) \right| \right| \leq E$$

where (x_0, y_0) represents current pixel coordinates, (x, y) represents neighbour pixel coordinates, E is a non-negative magnitude threshold.

3. Criterion 2: similar direction of the gradient operator.

$$\left| \alpha(x, y) - \alpha(x_0, y_0) \right| \leq A$$

where A is a non-negative angle threshold.

Local processing

4. Criterion 3: close distance between current and neighbour pixel points.

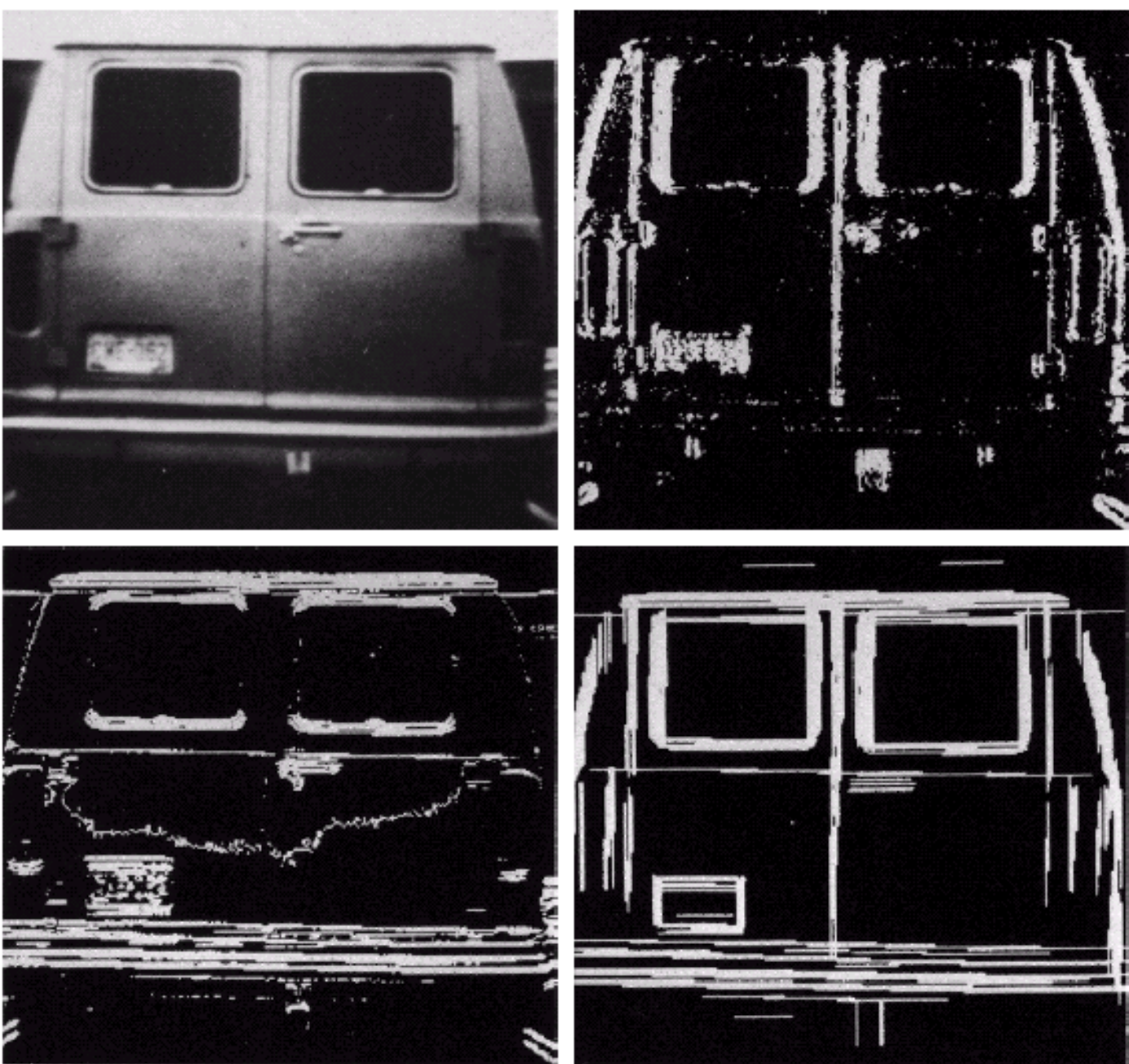
$$\sqrt{(x - x_0)^2 + (y - y_0)^2} \leq D$$

where (x_0, y_0) represents current pixel coordinates, (x, y) represents neighbour pixel coordinates, D is a non-negative distance threshold.

a b
c d

FIGURE 10.16

(a) Input image.
(b) G_y component
of the gradient.
(c) G_x component
of the gradient.
(d) Result of edge
linking. (Courtesy
of Perceptics
Corporation.)



$$E = 25,$$
$$A = 15^\circ$$

Global processing via Hough transform

1. Idea based on Hough transform: if two points lie on the same straight line, then they should have the same values of slope and y-intercepts.

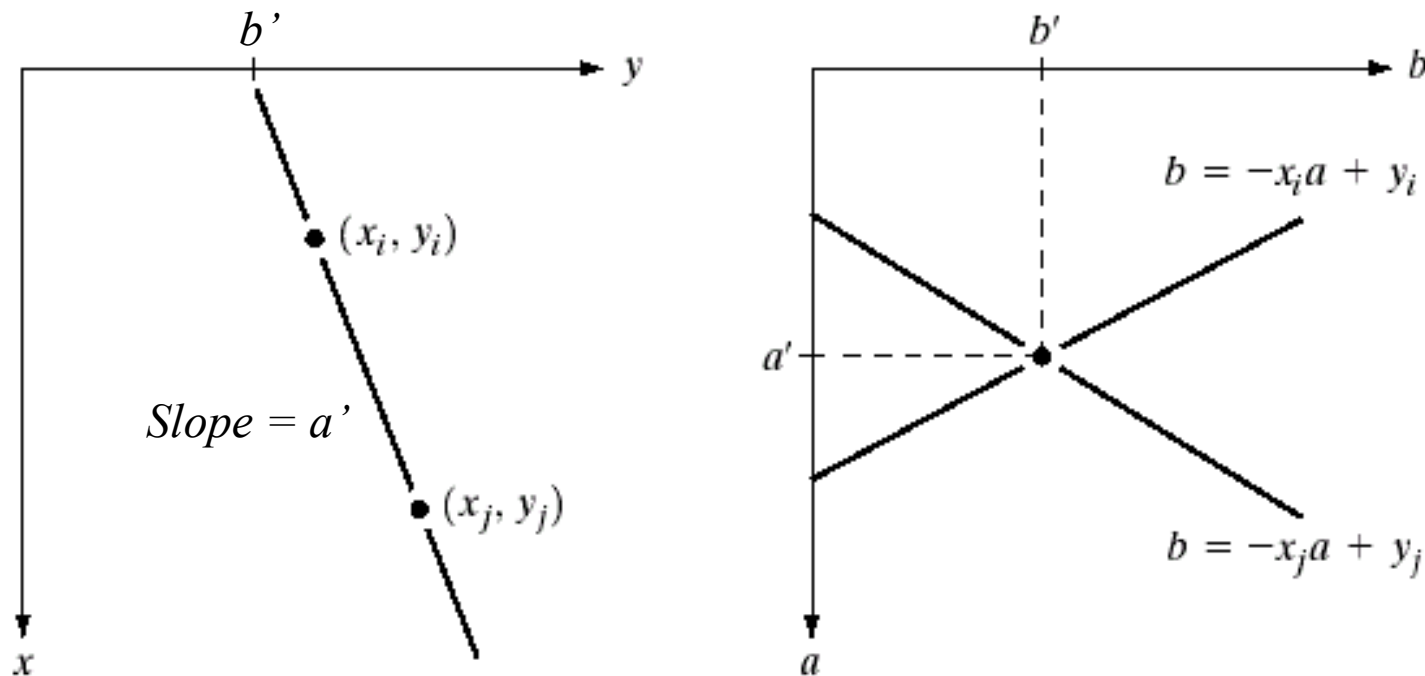


FIGURE 10.17
(a) xy -plane.
(b) Parameter space.

Global processing via Hough transform

2. Concepts and procedures:

- a. For a point (x_i, y_i) , we set up a straight line equation.

$$y_i = ax_i + b \Leftrightarrow b = (-x_i)a + y_i$$

where a = slope (variable), b = y-intercept (variable), x_i and y_i are known.

- b. We subdivide the a axis into K increments between $[a_{min}, a_{max}]$. For each increment of a , we evaluate the value of b .
- c. A relationship between a and b can be plotted in a parameter space.
- d. We partition the parameter space into a number of bins (accumulator cells), and increment the corresponding bin $A(a, b)$ by 1 (b is rounded into the nearest integer).

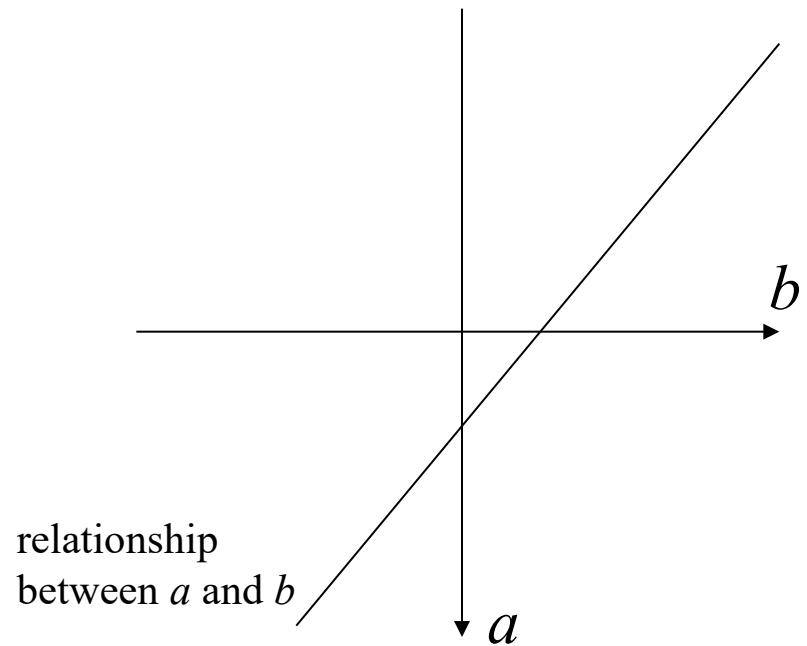
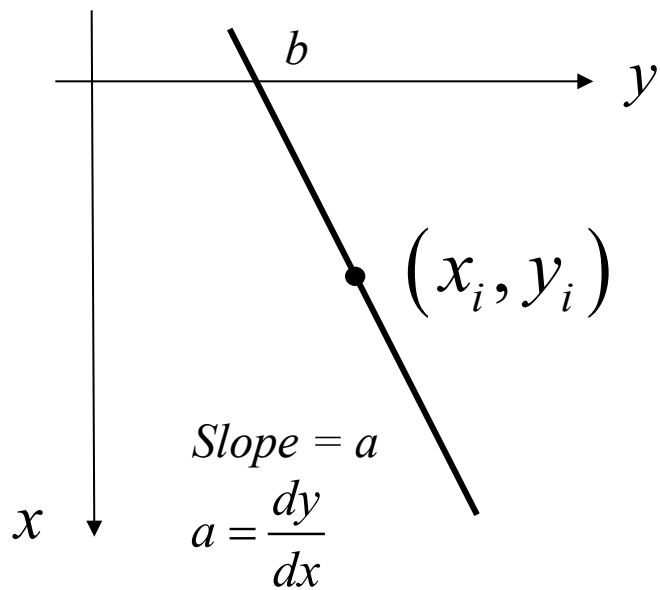
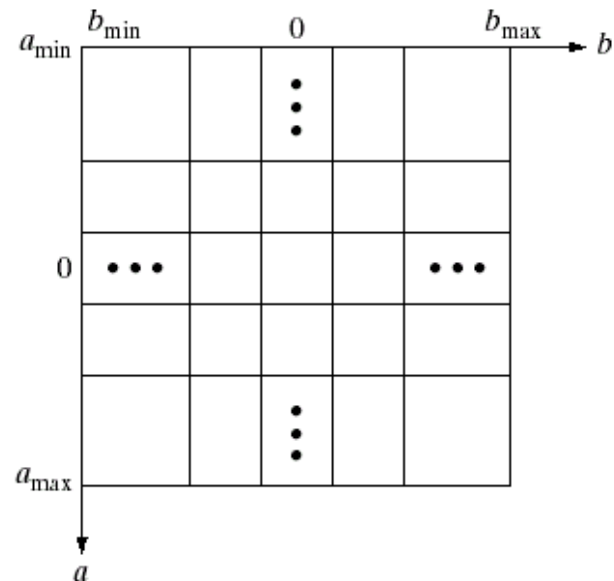


FIGURE 10.18
Subdivision of the
parameter plane
for use in the
Hough transform.

$A(a, b)$



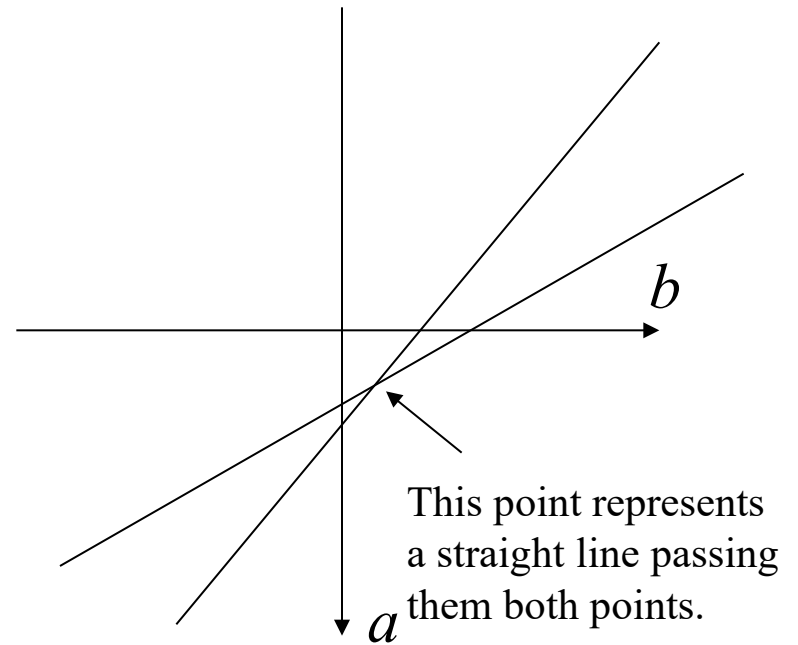
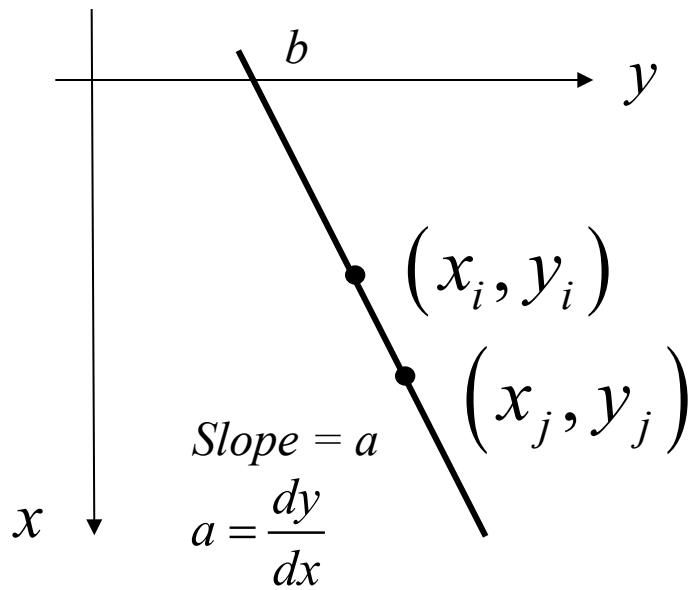
Global processing via Hough transform

2. Concepts and procedures:

- e. For another point (x_j, y_j) , we set up another straight line equation.

$$y_j = ax_j + b \Leftrightarrow b = (-x_j)a + y_j$$

- f. Similarly, we subdivide the a axis into K increments between $[a_{min}, a_{max}]$. For each increment of a , we evaluate the value of b . We plot the relationship between a and b in the same parameter space, and update bin values in the discrete parameter space.
- g. The bin $A(a, b)$ having the highest count corresponds to the straight line passing through the points (x_i, y_i) and (x_j, y_j) .

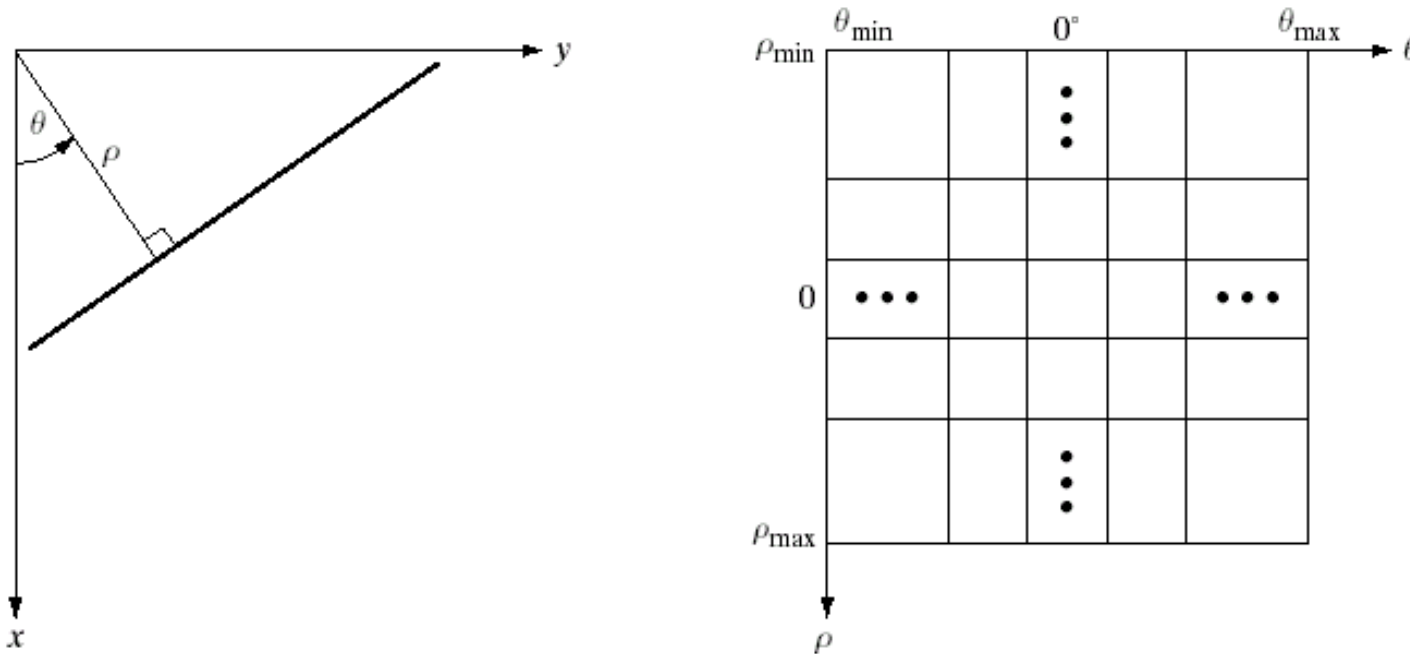


Global processing via Hough transform

2. Concepts and procedures:
 - h. The same procedure can be applied to all points. The bin $A(a,b)$ having the highest count corresponds to the straight line passing through (or passing near) the largest number of points.
3. Problem? Values of a and b run from negative infinity to positive infinity. We need infinite number of bins!

Global processing via Hough transform

4. Solution: use normal representation of a line.



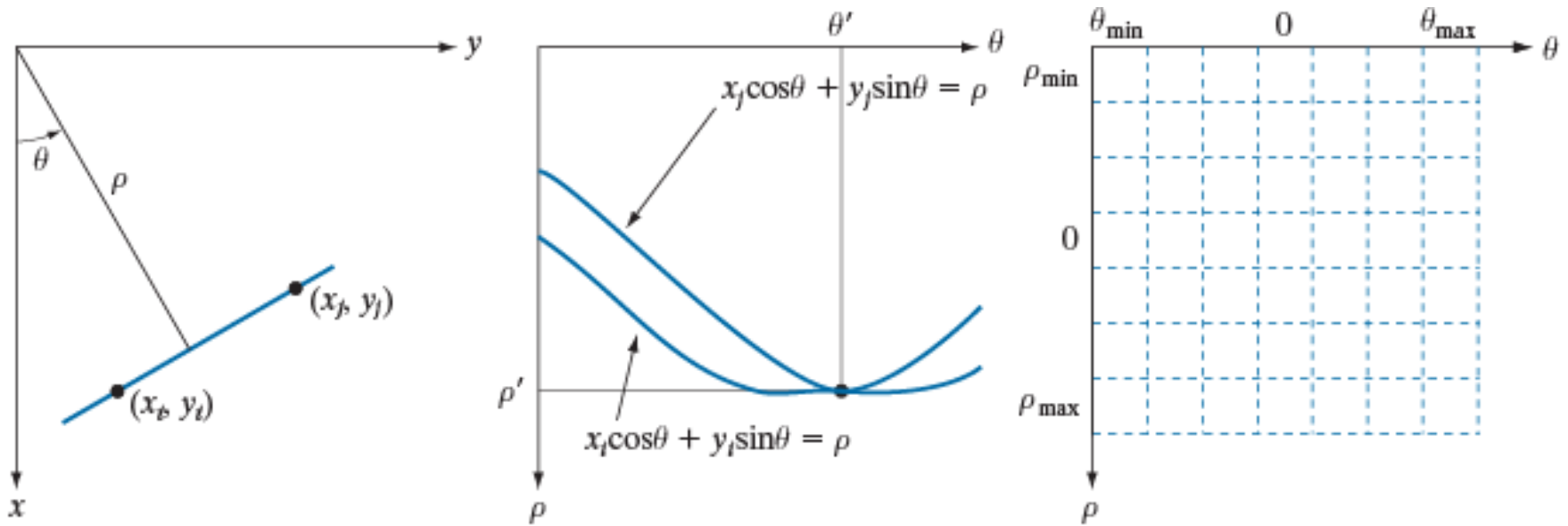
a b

FIGURE 10.19

(a) Normal representation of a line.
(b) Subdivision of the ρ - θ -plane into cells.

5. Theta runs from -90° to 90° . Rho runs from $-\sqrt{2}D$ to $\sqrt{2}D$, where D is the distance between corners in the image (length and width).

Global processing via Hough transform

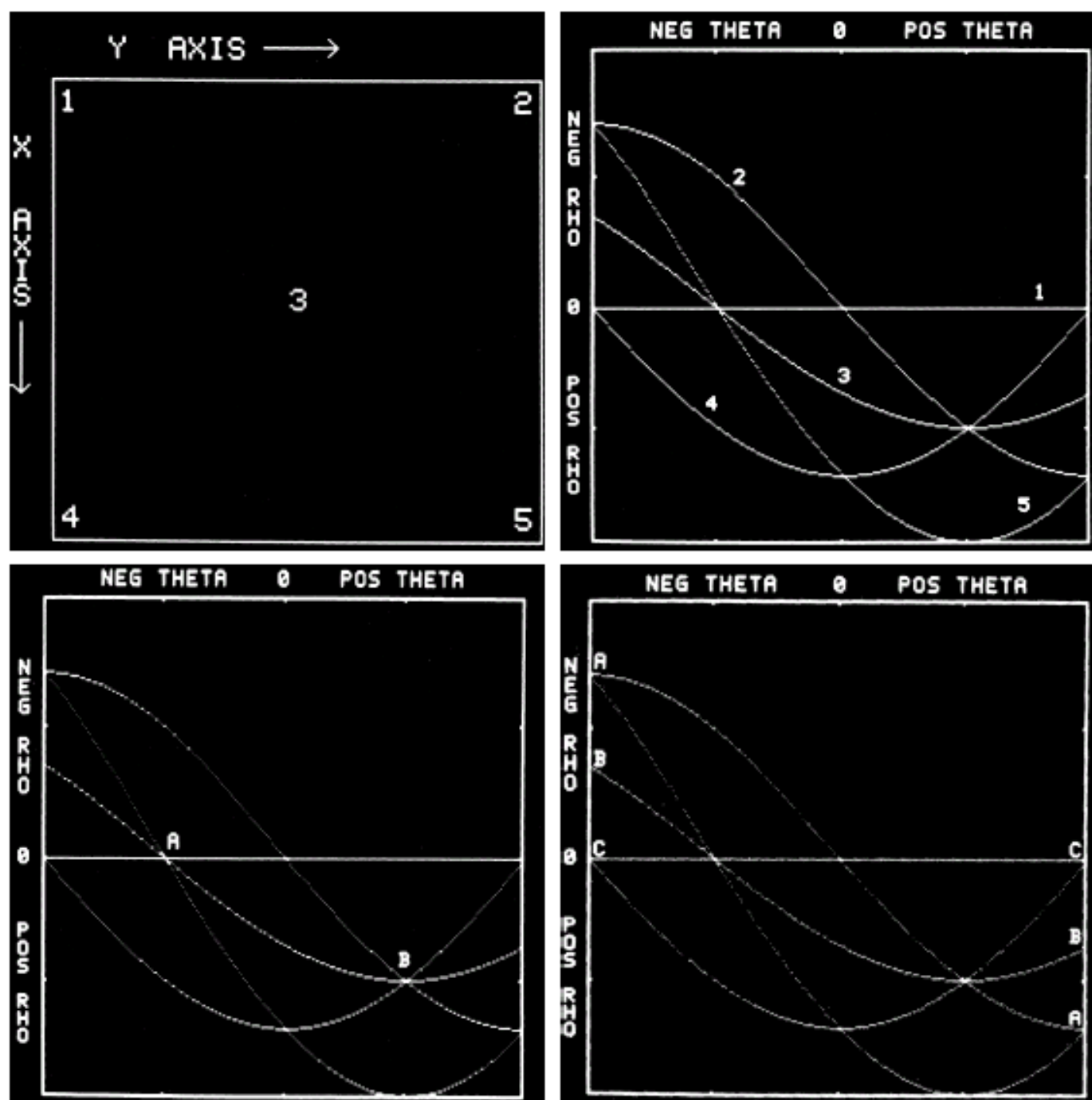


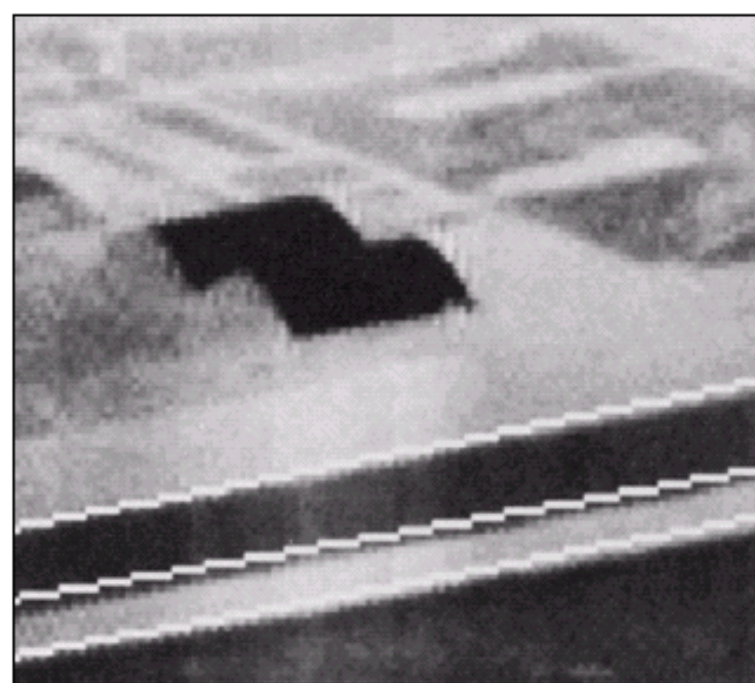
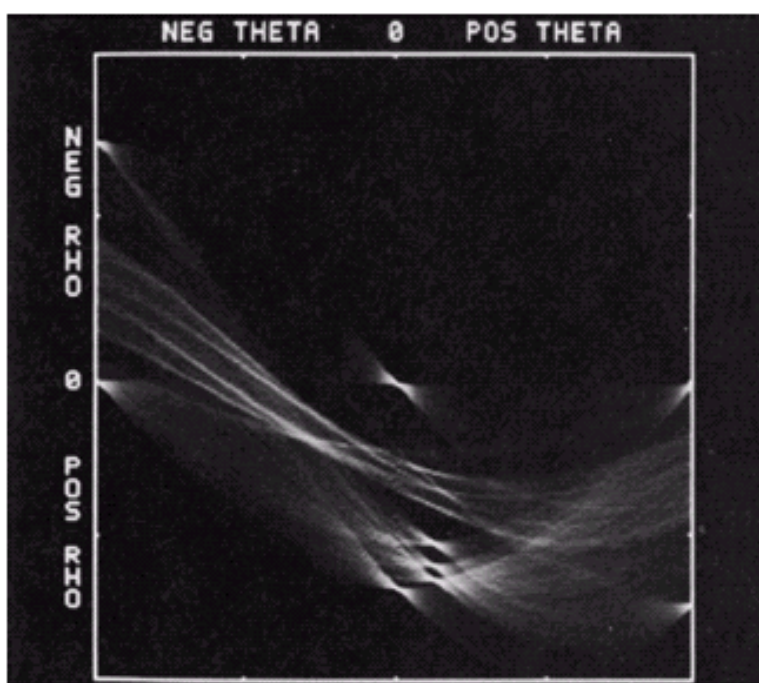
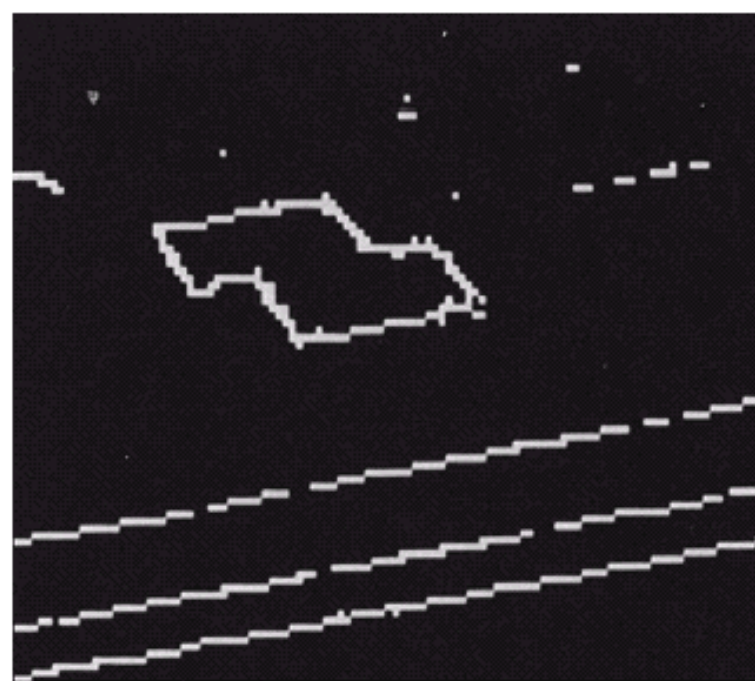
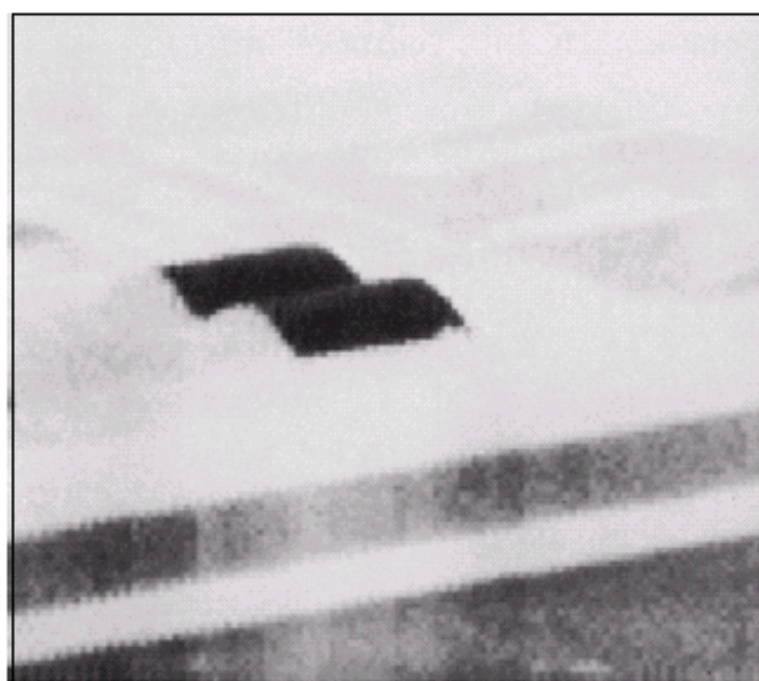
a b c

FIGURE 10.29 (a) (ρ, θ) parameterization of a line in the xy -plane. (b) Sinusoidal curves in the $\rho\theta$ -plane; the point of intersection (ρ', θ') corresponds to the line passing through points (x_i, y_i) and (x_j, y_j) in the xy -plane. (c) Division of the $\rho\theta$ -plane into accumulator cells.

a	b
c	d

FIGURE 10.20
Illustration of the
Hough transform.
(Courtesy of Mr.
D. R. Cate, Texas
Instruments, Inc.)





a	b
c	d

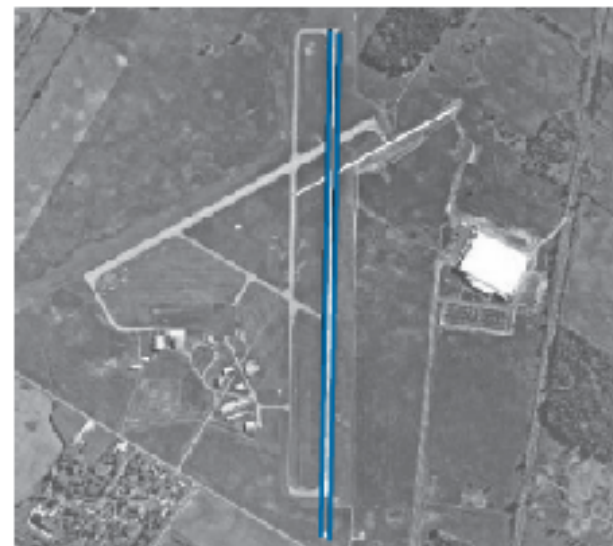
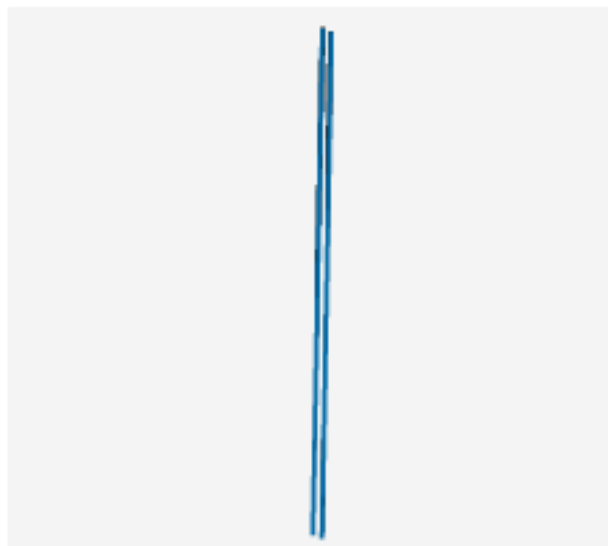
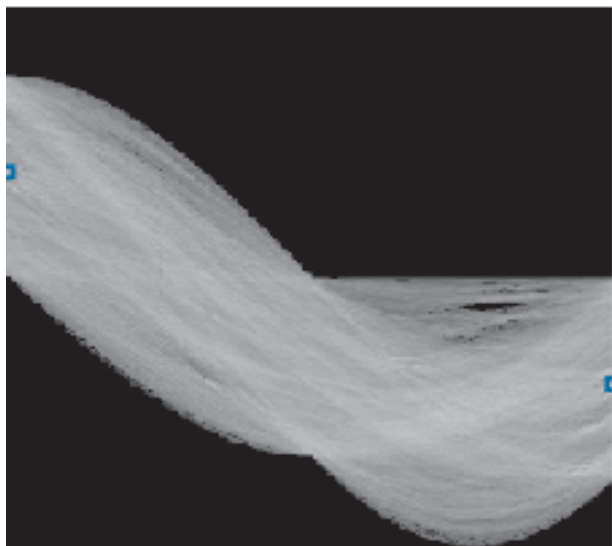
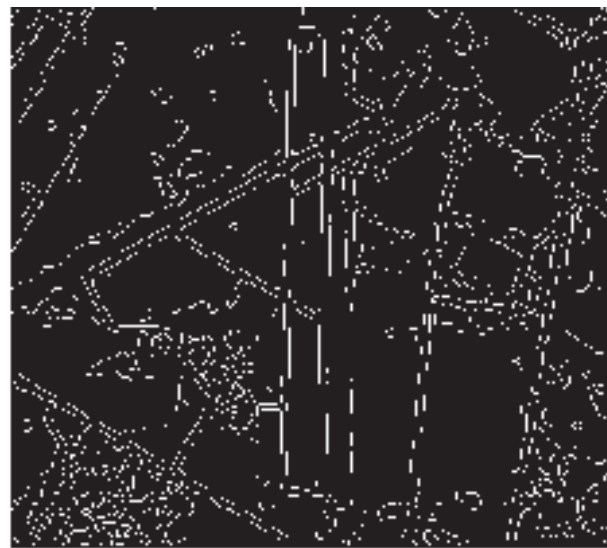
FIGURE 10.21

(a) Infrared image.

(b) Thresholded gradient image.

(c) Hough transform.

(d) Linked pixels.
(Courtesy of Mr. D. R. Cate, Texas Instruments, Inc.)



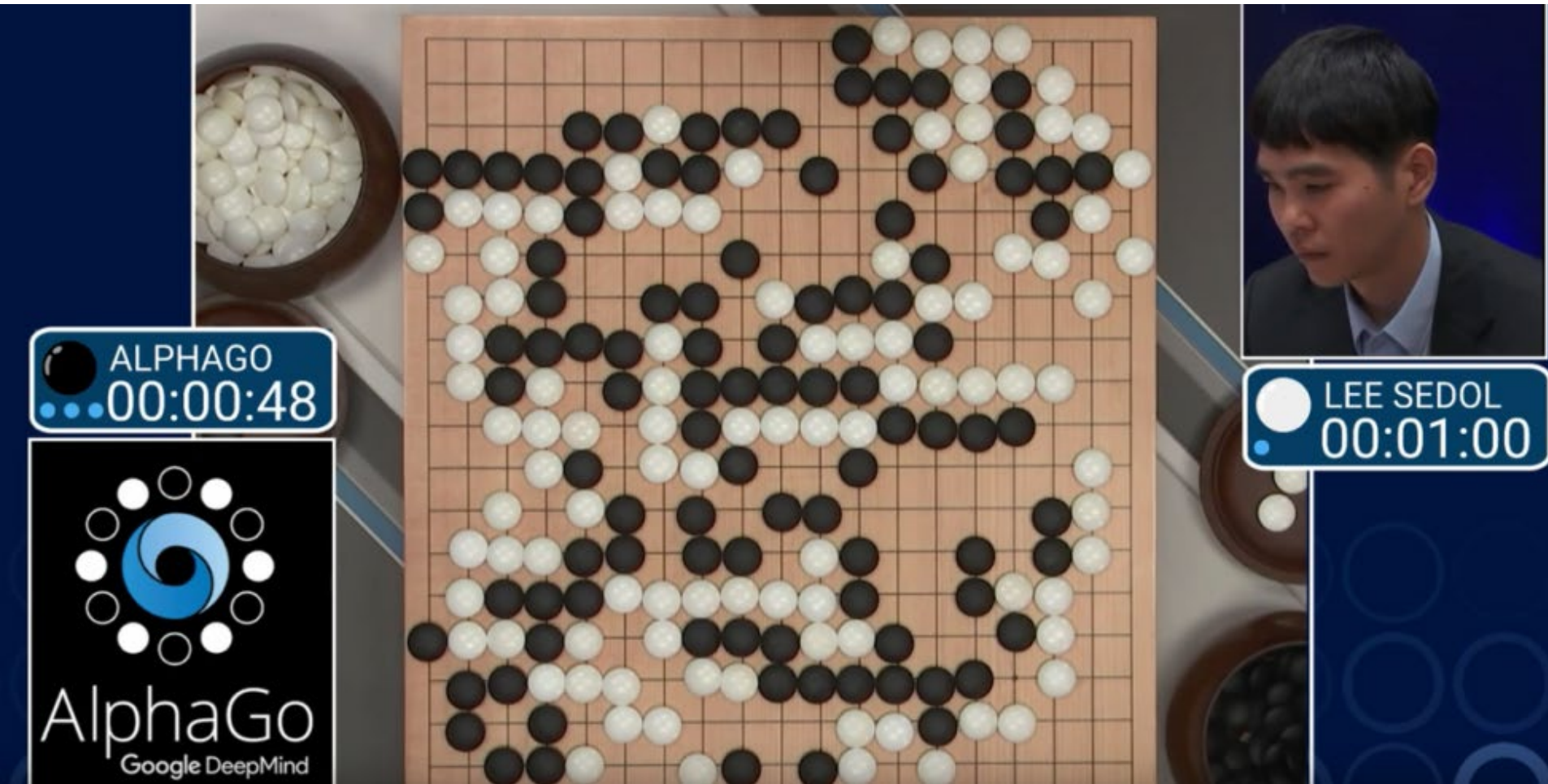
a b
c d e

FIGURE 10.31 (a) A 502×564 aerial image of an airport. (b) Edge map obtained using Canny's algorithm. (c) Hough parameter space (the boxes highlight the points associated with long vertical lines). (d) Lines in the image plane corresponding to the points highlighted by the boxes. (e) Lines superimposed on the original image.

Global processing via Hough transform

6. Solution to the edge linking problem:
 - a. Compute the gradient of an image and threshold it to obtain a binary image.
 - b. Partition the parameter space, e.g., a - b plane or theta-rho plane.
 - c. Examine the bin counts, pick the highest count.
 - d. Examine the relationship (based on criteria) between pixels.

Finding stones in a board of a 19x19 grid.....



Global processing via Hough transform

7. This method can also use a circle instead of a straight line.
How? (Discussion in class)

$$(x - c_1)^2 + (y - c_2)^2 = c_3$$